

Natural Interaction

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Chapter 1

Introduction

We mean by interaction the interaction between human and agent. Agent is an artificial agent, like chatbot, virtual agent, virtual reality, it could be a robot. We have the problem of beating up interaction between the two.

Usually we have some sort of interface while through exchange information, back in time the interface were mouse and keyboard, at the certain point technology gave us the opportunity of more sophisticated interfaces. the idea is to have kind of interaction between agent and human much like what we would have between humans. What is available of interaction is based on social signal, we might distinguish between verbal behaviour (in the sens of speech and in the way we communicate), behind the word that we are saying there is the sens of what we are saying. This give constraint of our interaction. Interaction is based also on gesture, mutual gaze, vocal behaviour, posture, obviously these are much conditioned by culture, that's made the very important problem of communication, it's that the person must interpret all this stuff in order to understand the theory of mind (I'm thinking that the other person think). Then might be context, contextual variables like height, beauty, placement in place (interpersonal distance). A number of complex factor that are going to modulate the social interaction.

The deal with agent is to module the agent to act like a human giving them some behaviour. In group there are more problems, like leadership and dominance. When we have a group of people interacting we are in a social psychologist problem. In this case we can e in a multi-agent environment, and it's a very complex system.

Then we might extend the non verbal behaviour and intersect with language, and here come the words. Nowadays we can use LLM, interact with voice, to interact and use these model to do something else than just generating stuff.

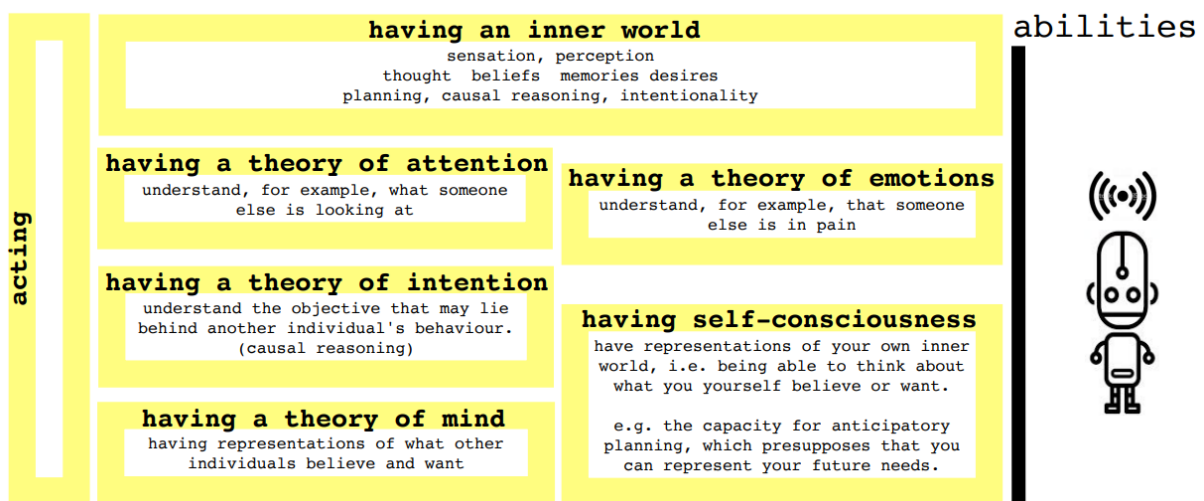
Further we'll involving in some of these problem dive in mind reading, literally using mind wave and also figuring understanding the behaviour of human.

Artificial Intelligence is used in order to understand the behaviour of human.

1.1 Social Signal Processing

There are number of components that may be used to achieve the full communication stack, that is a kind of recursive interaction because I know that you answer because I think that you think that I think etc.. Clearly in the evolutionary line lower level animal don't have this kind of theory of mind. But in order to achieve theory of mind we have kind of progression to basic capability to theory of mind. The basis is having an inner word (thinking, believing) there are classically associated this abilities with, for instance in CS with AI. Minimal cognition in word is what we refer to cognition, AI is a view an other is cognitive psychology, so when we have a nervous system (but even only perceptions, like in bacteria) we have an inner word. Then in order to arrive having a theory of mind evolution gone in other steps, one is having have a theory of attention, this mostly deal whit attention of objects an people, it's based on gaze, we can get a lot of information giving attention to people. Next step is theory of intention, we might figure out some causal reason why we have to focus our attention on certain object, in other word understand the objective that may lie behind another individual's behaviour (causal reasoning). Obviously we don't find this in animals, but there is a debate. Than to go further we may have sens of our intercepting world, our inner world that predents to the body, is loosely related of out emotion but certainly this is a kind of basis capability to

communicate with other, because having a theory of emotion means to understand the same emotion in others, this is the base for empathy (means to have a theory that, for instance, someone is in pain) we can understand the pain of people by inferring (posture..) or we have an other interpretation which is that of simulation, I understand someone is in pain because I perceive something (posture, vocalization...) and in some way I have a simulation in my body and I prove the same pain. Then we come to this two upper level, having a theory of mind, having a truly recursive theory (like said before having representations of what other individuals believe and want). In parallel we may have self-consciousness, we feel what we are, where, have representations of your own inner world, being able to think about what you yourself believe or want. It's necessary have a self-consciousness to have a theory of mind? good question, the two things may not depend one on other.



All this different level that we have in the full spectrum of human beings are putting in service of action, we are also lower level animal that are not evolved just to have pure thinking but our base goal is survive the world.

Social signal are something that we use for communicate, we may have some minimal form of communication (for example new borns).

1.1.1 Root of communications

At the roots we basically have some kind of vocalization and gestures. Obviously here we are not dealing with language but we have a form of vocalization which becomes in a more sophisticated way the prosa, modulating the voice help us to communicate emotions. Basically is kind of the three groups of signal that apes are able to produce:

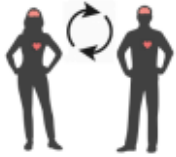
- Vocal displays
- Intention movements: mother may intend the new born was engaged by playing for instance
- Attention-getters: serves to draw attention to the face and posture display

Basic gestures like this is the initial way to communicate with robots. In the other side when we think of brain machine interface there are some kind of gestures. We are very skilled to communicate through gestures.

The basic message is that we start from this and could be also a choice of engineering perspective, we could start build basic layer and then build more complicated things.

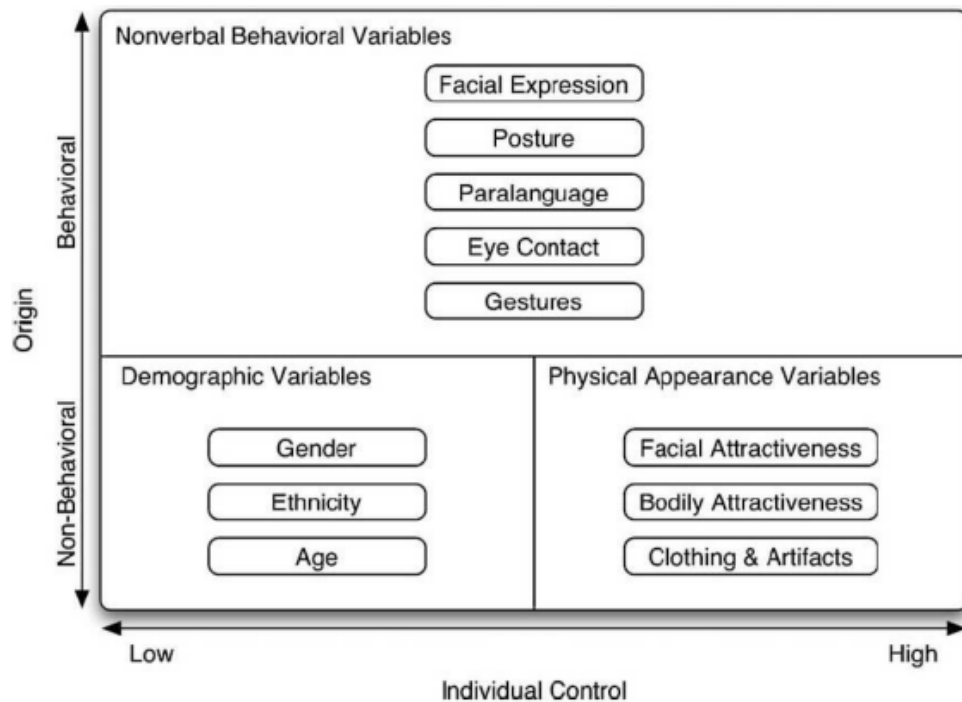
From an evolutionary perspective we need some sort of HW:

- pointing/gestures: emerged first in evolution (and individual ontogeny) in the natural, spontaneous gestures of pointing and pantomiming
- infrastructure of shared intentionality: in support of collaborative activities
- Conventional communication: language

			
	intentional communication	first glimmers of cooperative communication	recursivity = fully cooperative communication
communicative motifs	requesting	helping sharing	norms of cooperation
intentionality in communication	understanding goals		shared goals and communicative intentions
	understanding perceptions		joint attention and common ground
	practical reasoning		cooperative reasoning
communicative devices	ritualized signals	imitation	communicative conventions

1.1.2 Taxonomies

Usually signal that we use for communicate are defined by using taxonomies, there are several of this, we are using this one that is clear to understand in the purpose and in the use. We have two axes, on x we have the control (how much we control) obviously we can control clothing but we might have low control for our age or gender (now gender is under debate). On the y axes we have behavioural (what we said before) and the behavioural like facial expression, posture, gestures...



This might be interesting in some application exploiting machine learning and fashion, it's kind of fashion Netflix obviously we have to interpret the physical appearance.

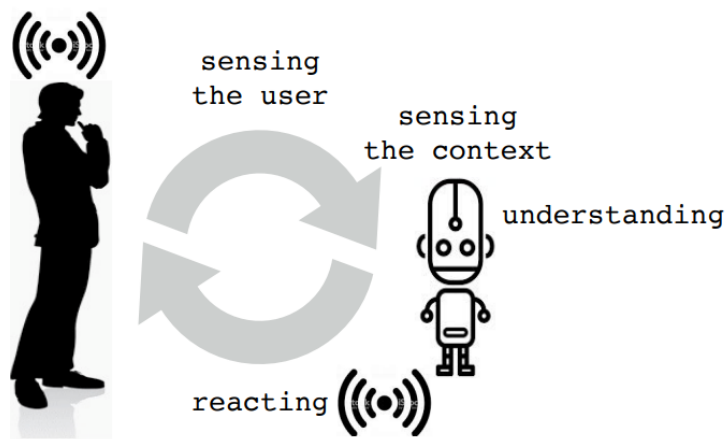
What kind of techniques are involved, in this matrix we can see which technologies are used for different kind of communication and we have different purposes for evaluating social behaviours:

	Example Social Behaviours							Tech.		
	emotion	personality	status	dominance	persuasion	regulation	rapport	speech analysis	computer vision	biometry
Social Cues										
Physical appearance										
height			✓	✓					✓	✓
attractiveness		✓	✓	✓	✓		✓		✓	✓
body shape		✓		✓					✓	✓
Gesture and posture										
hand gestures	✓	✓			✓	✓	✓		✓	✓
posture	✓	✓	✓	✓	✓	✓			✓	✓
walking		✓	✓	✓					✓	✓
Face and eyes behaviour										
facial expressions	✓	✓	✓	✓	✓	✓	✓		✓	✓
gaze behaviour	✓	✓	✓	✓	✓	✓	✓		✓	
focus of attention	✓	✓	✓	✓	✓	✓	✓		✓	
Vocal behaviour										
prosody	✓	✓		✓	✓		✓	✓		
turn taking	✓	✓	✓	✓		✓	✓	✓		
vocal outbursts	✓	✓		✓	✓	✓	✓	✓		
silence	✓		✓				✓	✓		
Space and Environment										
distance	✓	✓	✓		✓		✓		✓	
seating arrangement				✓	✓		✓		✓	

1.1.3 Systems

This is the basic scheme, we are trying to put together all this factors in order to capture the interaction loop:

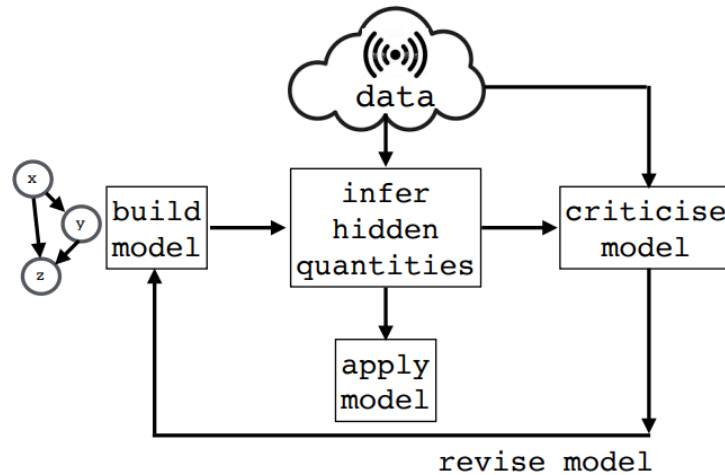
- sensing the user
- sensing the context
- understanding: in order to provide reaction
- reacting



1.1.4 Architectures

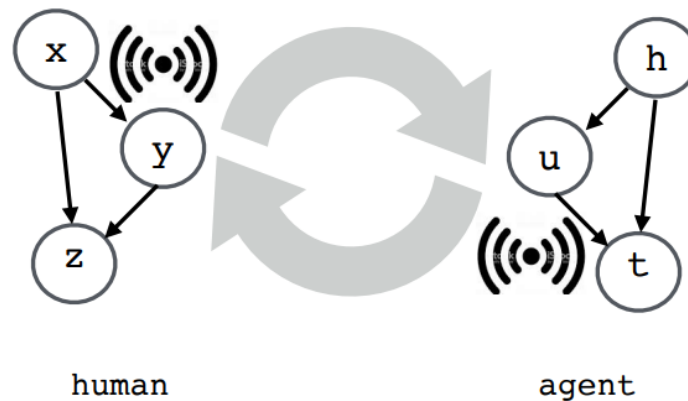
This is the complete pipeline:

- Data capture: we need to use some kind of sensors
- Person detection: detect the direction of the head, body or other cues size of the body, gender...
- Behavioural cue extraction: if you go through a classical pipeline you will rely on certain features, for instance classical gaze direction detectors are used in relay of high resolution face video
- Context understanding
- Social signal understanding
- Reacting: you might choose a policy to interact. In case of robotic it's a specific executable plan.



1.1.5 Models behind

As we told since we'll dealing with model, how do we model, in the higher level. We want some model of what is happening, here we are basic leaving in uncertain world, AI is uncertainty (may be noise or missing knowledge of worlds).



Best way to represent probabilistic model is using probabilistic graphical modeling, we have this graph and the node is equal to a random variable, the arrow is conditional dependencies between variables, so basically this graphical model is modelling the joint probability of $P(x, y, z) = P(z|y, x) \cdot P(y|x) \cdot P(x)$ (probability of z given y and x times probability of y given x times probability of x). This as we'll see is a generative model, is a generative because if I know this distribution I can instantiate data from this.

This is the process we'll go through, each model we have parameters (like mean, standard deviation...) and we have to learn this parameters (in machine learning) or infer it in statistical. Then once we have learned our model we have to test it and see what happens. If it's not ok we'll go through a revision of model.

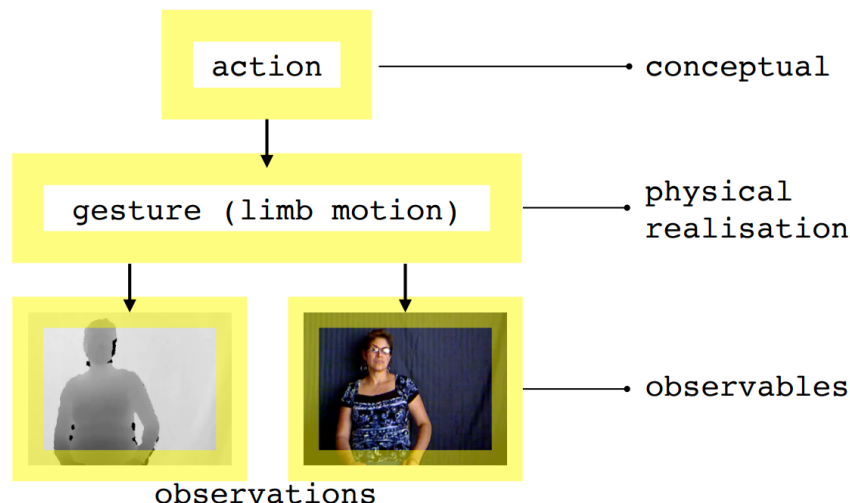
Clearly building the model will deal both with psychology and neuroscience.

Chapter 2

Theoretical Models

PGM provide a suitable, in general, tool.

2.1 Theoretical model for gesture



Understand gesture is a typical problem of natural interaction, because we want to use gesture to interact with system, like robots and stuff like that. In the most simple case these gesture describe an alphabet. In general this gesture is an action, action means that if you think the way you perform an action we may consider an action as kind of concept (conceptual representation in our mind). We have hierarchy of gesture usually. Generally we don't want to make a agent perform an action but we want that the agent infer the gesture (inverse probability model) we have the observable and what we want to do is going back to conceptual action. The full model has a kind of two ways, generative model is exactly what we do in our life, we have in mind a concept that we translate in movement.

Inference is possible if I have a generative model, in simple case means have available the way I reason and have the projection of our reasoning. Not always is possible to have a full generative model, what happens if we don't have this model? the problem that we want to solve is still the inverse problem, in that case we might do some simple tricks in which we can implement this part, these are called discriminative model, we have feed forward neural network (CNN) and we can build a classifier and given the input data it'll output the action labelled.

Basically this scheme is kind of abstraction:

- Action: might be a describe concept, binary stuff in this case, thumb up or down
- Gesture: is going to implement the action.
- Observation: now we are in 3D space, we have sensors and we know from computer vision that if I show to a camera a movement this movement is translated in a certain frame that are pixels settings.

Now we want to summarize this in the higher level, using the Probabilistic Graphical Models. To do that the first step is identifying random variable, we have some kind of

outcome in the event space, and we assign a value. Means identifying the basis of the model, in the previous example we had:

- Action: let's call a random variable $A = \{up, down\} = \{0, 1\}$ binary
- Gesture: can be much more complicated, in robotics we can have joint angle of what we are moving, the variable G is a time dependent random vector which components are the variation of angles in time: $G(t) = [\alpha_1(t), \alpha_2(t), \alpha_3(t)]$
- Depth: is composed by gray level images so we'll have sequence of frames. Variable D is a time dependent random matrix
- Video: is even more complicated, we have groups of frame. Variable V is random tensor: $V_{RGB}(t)$. Each tensor is a RGB frame, one frame each.

Even in this simple case we have find a lot of variables. We have the description of a problem given by description of these random variable, the more complete model (full information) we'll have if we have the joint probability of these variables:

$$P(A, G, D, V) = P(A \cap G \cap D \cap V)$$

To represent the joint probability we have to use a full information matrix from which we can derive all the possible states, this give us FULL INFORMATION:

P(A, G, D, V)	A	G	D	V
0.3	0	0	0	0
0.01	0	0	0	1
...	0	0	1	0
...			...	
...	1	1	1	1

We note that P is the summation of all the probability in every state (sum to 1) and in this case we found a 2^4 matrix.

2.2 Joint probability factorization

It's based on the product rule: $P(A, B) = P(A)P(B|A)$. We can apply this rule to the joint probability and find:

$$P(A, G, D, V) = P(A|G, D, V)P(G|D, V)P(D|V)P(V)$$

We know that all factors are tables, note that we need to write only one table because the other is given from free using the complement rule (1 - table).

$P(A = 1 G, D, V)$	G, D, V
0.1	0 0 0
0.01	0 0 1
...	0 1 0
...	...

Table 2.1: Factor 1 = $P(A|G, D, V) \rightarrow 2^3 = 8$

$P(G D, V)$	D, V
...	0 0
...	0 1
...	0 1
...	1 1

Table 2.2: Factor 2 = $P(G|D, V) \rightarrow 2^2 = 4$

$P(D V)$	V
0.3	0
0.7	1

Table 2.3: Factor 3 = $P(D|V) \rightarrow 2^1 = 2$

The last factor is a 2^0 matrix, so in total we found 15 row matrix.

When we factorize the advantage is that factors are easier to learn, because we can learn these table in a separate way. It's a kind of divide et impera approach, and we know that we can go back to the full table multiplying all the factor. Basically the dimensionality are the same of the full table.

The question now is how do we simplify?

There is key concept of probability that is independence, we have two kind of independence:

- marginal independence: statistical independence, we have two random variable x, y and we say that they are independent when we have this kind factorization $P(X, Y) = P(X)P(Y)$. Also known as statistical independent.
- conditional independence: we have three random variables and we say that X is conditional independent $\perp$ of Y given Z if the following holds $P(X, Y|Z) = P(X|Z)P(Y|Z)$. It's not only a mathematical fact, but conditioning the probability of X means that we know the probability of certain random variable after we observe something on the random variable that is correlated (Z in this case)

Let's see an example, first of all we using product rule to divide the joint probability:

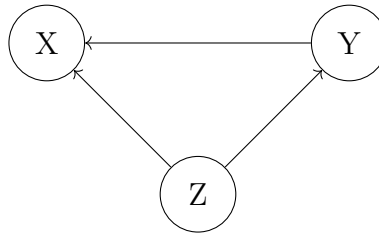
$$P(X, Y, Z) = P(X|Y, Z)P(Y|Z)P(Z)$$

$2^2 \quad 2^1 \quad 2^0$

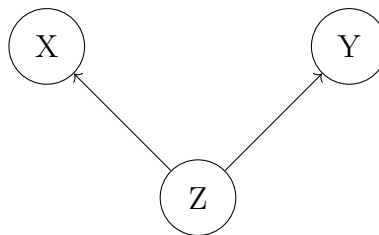
Then we do an assumption, for some reason, we say that X is conditional independent of Y $X \perp Y$, in this case we can simplify the probability of the factors, in this way we are reducing dimensions (we are reducing the lines of tables). We obtain:

$$P(X|Z)P(Y|Z)P(Z)$$

We can visualize thought PGM the formula of joint probability and its factorization, nodes are random variables and arrows are conditional dependence:



Given the assumption happens that we simplified the graph:



Remember that arrow direction depends on decomposition and should make sense for the problem. This is said here because we've seen the example of the probability to get a grade given a study process and the intelligence, probability of grade depends on intelligence make sense but the reverse doesn't.

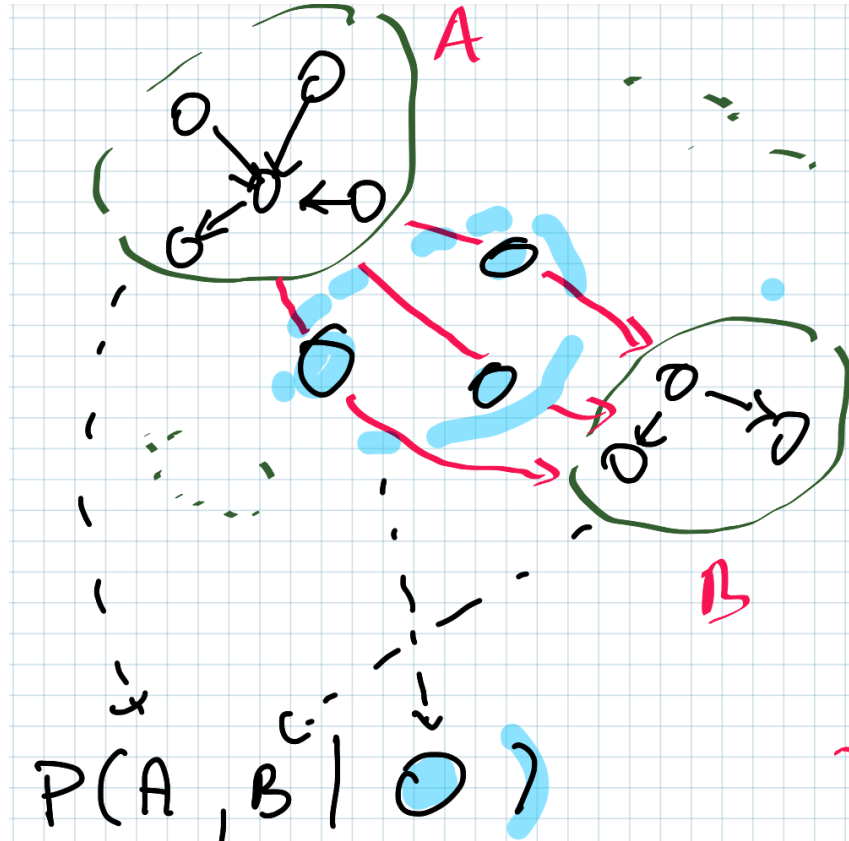
2.3 Variable independence

Given a variable in PGM's we might determine if this is independent from an other. From a PGM's we can set up probabilistic query, that is for instance the probability of V_j , or the probability of V_j, V_k .

A way to seeing this graph is considering this graph like a knowledge graph which each node represent concepts in reality and we can filter some parts of this graph in order to concentrate in some property, and this property is expressed in probabilistic. The query can be seen as query in graph DB.

What we need to set up the graph is independent relations, if we are considering the certain path, the idea is that v_j is independent from v_k if for all the path that I may find to go from the two nodes, that pats are blocked:

We can extend this to groups of nodes, given two groups A and B and given n observed node, if all the paths that start from A and arrive to B pass through observed nodes are blocked then A and B are d-separated.



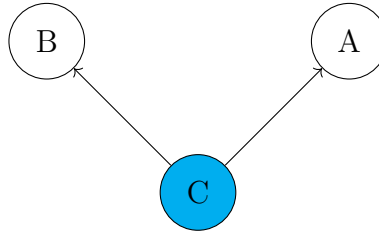
That means that the probability of group A and B are independent given the observed, and again we can decompose. Again if we can simplify like that the joint distribution we'll have simple tables:

$$P(A, B|O) = P(A|O)P(B|O)$$

Now let's see the condition, are three very simple condition to check the separation. Than we'll see that such condition can be used to simply express some fundamental ideas in statistics (machine learning), we'll see two example, the i.id problem and the Markov property.

2.3.1 Tail-to-Tail

The first condition is the **Tail-to-Tail** configuration, we might have a mini graph and this configuration, tail to tail because the node C has two tail, tail is the start of the arrow (the end is the head):



We have a structure, and in order to factorize we have to do what we learned before, we'll not use the full product rule but we'll use the structure that we've designed:

$$P(A, B, C) = P(A|C)P(B|C)P(C)$$

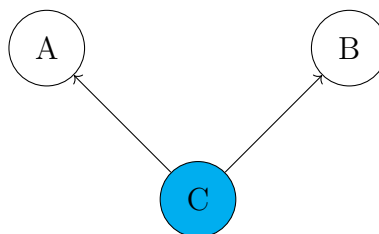
The general goal is to infer some conclusion on how A and B are related $P(A, B|...)$? We can distinguish in two cases:

- No nodes is observed, C is not observed
- C is observed, then we don't need to know B and A

In the first case we have to find some conclusion on the joint probability of A and B, but in this case we have the probability of everything $P(A, B)$ and we can use marginalization to infer some properties on C. Marginalization is the sum of all the possible value of C if variables are discrete (C is the income monthly), in a continuous case we'll integrate the values of C (C is the range of income), $\sum \int P(A, B, C)$. What we sum (or integrate) is the full probability, all the cases, again because C is not observed, we don't know anything about it. Assume discrete, we know the joint distribution because we know the structure and we can decompose, we find C in every factor and we can derive that, in this configuration there is no marginal independence. This means that not given the value of C, A and B become dependent because C become a latent factor and we have to marginalize:

$$\begin{aligned}
 P(A, B) &= \sum_C P(A, B, C) \\
 &= \sum_C P(A|C)P(B|C)P(C) \\
 &\neq P(A)P(B)
 \end{aligned}$$

Now consider the second case, in this case C is observed (for instance we know the income, that is high):

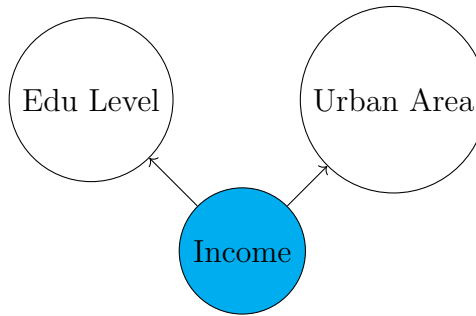


in this case we can calculate $P(A, B|C)$, we have conditional dependence because we observe C and we don't have to marginalize. Then we can use the conditional probability definition:

$$\begin{aligned}
 P(A, B|C) &= \frac{P(A, B, C)}{P(C)} \\
 &= \frac{P(A|C)P(B|C)P(C)}{P(C)} \\
 &= P(A|C)P(B|C) \\
 A &\perp B | C
 \end{aligned}$$

after all we can say that A and B are conditionally independent. The path from A to B is blocked, that means that C is going to block the information, because if we know C we can infer conclusion on B without knowing A (and vice versa), so this path is d-separated.

The example was the income of the family in the node C, than the education level in the node A and then urban living area in node B, so we have concrete concepts:



i.i.d concept

We have a fundamental concept in statistic, is the **i.i.d**, independent and identical distributed. All basic classical statistics is based on this assumption (and of course machine learning). Independent means that all observation are independent, identical means that we have a distribution of some parameters and they will keep during the experiment.

We'll see a simple example to demonstrate this rule, we start from the Bernoulli distribution, that model the throw of a coin. In this example we have two events, tails or head $t, h = S$, these are modelled by one random variable $X = 0, 1$. Then we can compute the probability of x given the random variable:

$$P_x(X = x) = \text{Bern}(x|\theta) = \theta^x(1 - \theta)^{1-x}$$

In the distribution we have theta parameter, that describe the probability, in this case 0 or 1, but what is the parameter measuring? the fairness of the coin, if theta is 0.5 means that we have 50% of each result, if theta is 1 means that we have 100% probability to have a certain result.

Then we have machine learning, that is something that we'll give the data and we have to learn the model. In statistical point of view we have to estimate the theta (confidence intervals) on other way (machine learning way) is maximum likelihood estimation (minimize the cost function, error function).

The likelihood is the probability of the observed data, are the joint probability of all the data that we have. Now the joint probability life in assumption that is the i.i.d, we are making this assumption because that means that the probability of all the observation x_n is the product of all the probability of x_i (in our case every observation is modeled by Bernoulli):

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n x_i$$

$$P(x_1)P(x_2)\dots P(x_n)$$

The likelihood function hold a parameter theta, is a function of the theta because we are using to determine the theta:

$$L(\theta|D) = P(x_1\dots x_n|\theta)$$

$$\prod_{i=1}^n P(x_i|\theta)$$

$$\prod_{i=1}^n \text{Bern}(x_i|\theta)$$

$$\prod_{i=1}^n \theta^{x_i}(1 - \theta)^{1-x_i}$$

In statistic we call this MLE, Maximum Likelihood Estimation, that means that we are going to determine and estimate theta, that is the parameter of the network. We have to find the theta which maximize the likelihood function, to maximize a function we have to differentiate, we differentiate the likelihood function and set to zero:

$$\theta_{MLE} = \text{argmax} L(\theta|D) = \frac{\partial L(\theta)}{\partial \theta} = 0$$

In Neural Network we compute derivatives along the network is done by back-propagation and using the technique of gradient descent because there are some case in which we cannot compute derivatives locally.

Given the likelihood function is convenient to have the log of the function because we have exponents, when we have exponent distribution it's convenient use the log because log is monotone:

$$\log - \text{likelihood} = \sum_{i=1}^n x_i \cdot \log \theta + \sum_{i=1}^n (1 - x_i) \log(1 - \theta)$$

$$= \log \theta \sum_{i=1}^n x_i + \log(1 - \theta) \sum_{i=1}^n (1 - x_i)$$

Now we want to determine the derivative in respect to theta. End than I'm gradient descent equals zero:

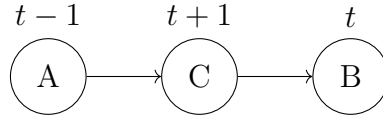
$$\begin{aligned}
\frac{\partial l(\theta)}{\partial \theta} &= \frac{1}{\theta} \sum_{i=1}^n x_i - \frac{1}{1-\theta} \sum_{i=1}^n (1-x_i) = 0 \\
\frac{1}{\theta} \sum_{i=1}^n x_i &= \frac{1}{1-\theta} \left[\sum_{i=1}^n 1 - \sum_{i=1}^n x_i \right] \\
\frac{(1-\theta)}{\theta} \sum_{i=1}^n x_i &= n - \sum_{i=1}^n x_i \\
\frac{1}{\theta} \sum_{i=1}^n x_i - \sum_{i=1}^n x_i &= n - \sum_{i=1}^n x_i \\
\theta_{MLE} &= \frac{1}{n} \sum_{i=1}^n x_i = \frac{6}{12} = 0.5
\end{aligned}$$

In the end we found the mean, we sum over the head (in our example the number of head) and then divide by number of data, in our case we obtain: $\frac{6}{12} = 0.5$

This is the MLE, we have to learn the theta. We can do this with all the distribution, for example for the Gaussian.

2.3.2 Head-to-tail

It's the other chain, it's very typically and it's called Markov chain. We are considering the case when the random variables are not observed, and we want to infer some conditional probability of A and B.



Again what we want to know is the probability of A and B:

$$P(A, B, C) = P(A)P(C|A)P(B|C)$$

We are marginalizing for all the states of C

$$\begin{aligned}
P(A, B) &= \sum_C P(A, B, C)^* \\
&= \sum_C P(A)P(C|A)P(B|C) \\
&= P(A) \sum_c P(B|C)P(C|A) \\
&= P(A) - P(B|A) \\
&\neq P(A) \cdot P(B) \\
P(A, B) &\neq P(A)P(B) \\
&\neg A \perp B
\end{aligned}$$

This means that A and B are not marginal independent.

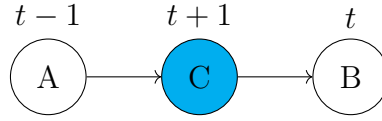
*

$$C = 0, 1$$

$$\sum_{C \in \{0,1\}} P(A, B, C) = P(A, B, C = 0) + P(A, B, C = 1)$$

This means that there is no blockage from the C variable, A is transmitting information to variable B.

Now we consider the other case in which we are observing C, in order to see what are the variables A and B we want to consider what happens to the joint distribution, what kind of joint information they are sharing while I'm observing C:



$$\begin{aligned}
 P(A, B | \textcolor{blue}{C}) &= \frac{P(A, B, C)}{P(C)} \\
 &= \frac{P(A)P(C|A)P(B|C)}{P(C)} \\
 \text{By Bayes Rules } \frac{P(A)P(C|A)}{P(C)} &\rightarrow P(A|C) = \frac{P(C|A)P(A)}{P(C)} \\
 P(A | \textcolor{blue}{C}) \cdot P(B | \textcolor{blue}{C}) &\rightarrow A \perp B | \textcolor{blue}{C}
 \end{aligned}$$

This means that the path from A to B is blocked, there is not transfer of information. What does blocked means? in such a structure if I know C I don't know about A because we can infer from B. Assume that we are tracking, there is a camera of videosorveglianza system, I can determine future position of people because we infer some information about what we've seen. An other famous problem is language modelling.

Markov:

$$P(x_t | X_{t-1} x_{t-2} x_{t-3} \dots x_0) = P(x_t | x_{t-1})$$

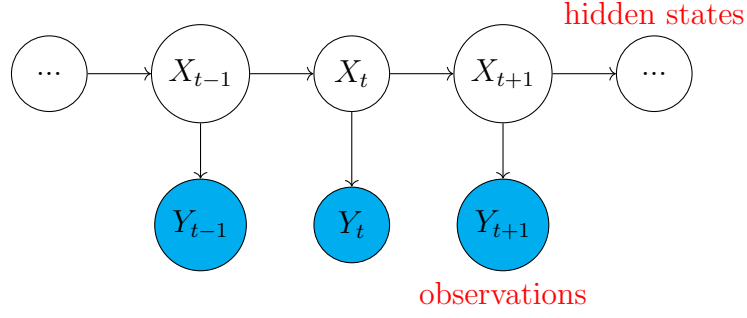
Means that we can assume that the probability of x_t is given by the observation at time $t-1$, and not all the previous observation.

There are some case in which we can not use this Markov assumption, for instance we have to complete this sentence: "I(x_0) have(x_1) seen(x_2) a(x_3) dog(x_4) baiting(x_5) his(x_6)(x_7)", we can consider all the previous words like a x observation, if we give from grant Markov we know that we can predict the missing word given only the previous observation, it's impossible, we need some other relationship:

$$P(x_7 | x_6) = P(x_7 = ? | x_6 = his)$$

We can model this problem with an auto-regressive model and not Markov model.

This model is very important in each dynamic model, like walking, gesture and so one, everything that I can track in time. The most simple model is the Kalman Filter, is a special case of this graphical model:



Our variable x_t could be a vector that can contain position, velocity, acceleration. Assume that there is no acceleration and my velocity is constant, now obviously that are the condition I don't need the Kalman filter we can just use Newton law, but the problem is that we are observing an object with certain mass and there might be noise (illumination condition, analog-digital conversion, camera noise). The other problem is the tracking, we never deal with the real state, but we have observation and this observation also have noise inside. So what we are observing is not the x , but is the variable y (the observations), the x 's are hidden states. So the actual problem is to infer the actual state because the noise might be very high.

Actually from a probabilistic point of view we have a joint probability pdf (probability density function) in which we have all the observables and all the states. If we use classic rules of probability we don't get out of the problem because we have too many factors, but we might say that if I arrive at a certain point and I've observed a certain state I have more probability to be in a certain state, so we don't need to use all the story of observation, we are simplifying the joint probability:

$$P() = P(x_1) \prod_{t=2}^t P(x_t|x_{t-1}) \text{dynamics model} \cdot P(y_t|x_t) \text{observation model}$$

In order to compute this I just need to think the dynamics model that is a Markov model. So we know if we have this model we have two fundamental models:

We can assume that the transition is Gaussian and also we assume that the observation noise is normal, in this case we have Kalman filter. We can also express this model with a dynamic equation:

$$x_t = \alpha x_{t-1} + \Sigma \rightarrow \Sigma \sim (\text{sampled}) N(0, 1)$$

and there is also the observation equation:

$$y_t = c \Sigma x_t + \Sigma_t \rightarrow \Sigma_t(t) \text{Gaussian perturbation} \sim N(0, 1)$$

How can we implement this? if we have data recurrent neural network, we don't even need to implement the Markov sum. Or what is done nowadays is using the transformers.

An other model, assume that we still have continuous output and we can use Gaussian noise, now we might consider x discrete, so for instance x might be {sunny, rainy, foggy}. So we have discrete transition of the states, so we can have a transition table with probabilities and a point for instance is the probability in a sunny day to stay in a sunny day. Observation could be meteorological observation of a camera. This model here is

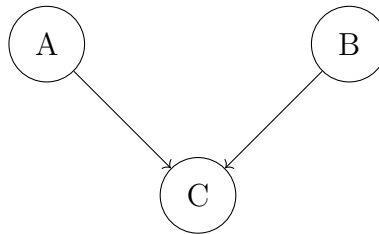
called the Hidden Markov Model, so the same structure, the same assumptions (Markov assumption) but here we are considering the hidden variable are discrete. This model generate the speech recognition, in speech recognition we have an ideal state (the x) that for instance could be B and when we hear we can hear a lot of noise.

2.3.3 Collider

Now we have the last model, it's called Collider (or Head to Head). A and B are kind of cause of C. This is an important structure, the behaviour of this structure is the opposite of the other 2, if I'm observing C, A and B will not be independent (observation will not brake the path).

$$P(A, B, C) = P(C|A, B)P(A)P(B)$$

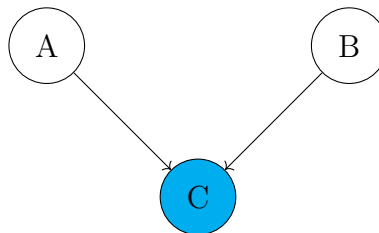
We are now considering the case in which C is not observed:



so again we have to pass through the marginalization:

$$\begin{aligned} P(A, B) &= \sum_C P(C|A, B)P(A)P(B) \\ &= P(A)P(B) \sum_C P(C|A, B) \rightarrow 1 \\ &= P(A)P(B) \\ A &\perp B \end{aligned}$$

A and B are marginally independent, exactly the opposite of the other cases. Now the case in which we observing C:

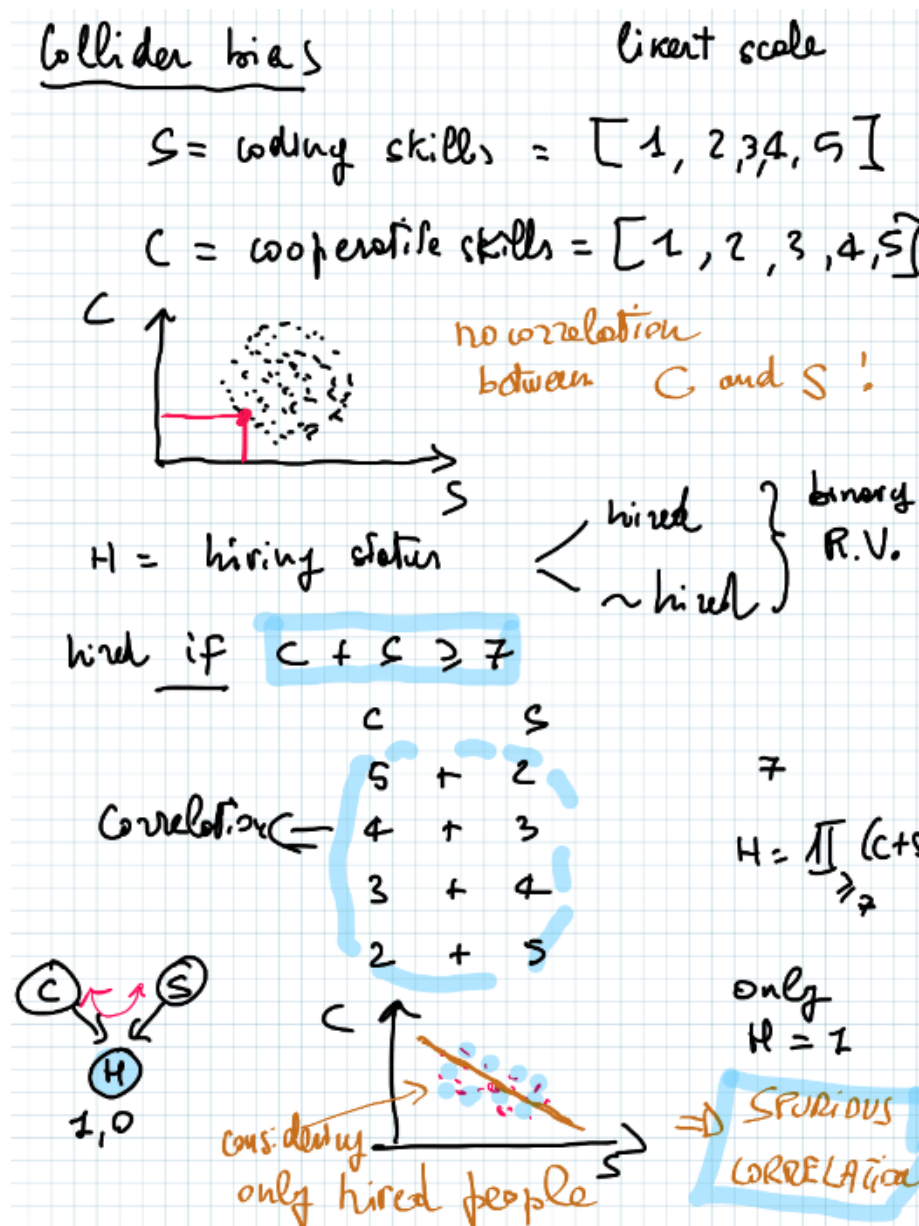


$$\begin{aligned} P(A, B|C) &= \frac{P(A, B, C)}{P(C)} \text{ definition of conditional probability} \\ &= \frac{P(C|A, B)P(A)P(B)}{P(C)} \\ &\neq P(A|C)P(B|C) \end{aligned}$$

This means that they are not conditionally independent.

Collider Bias

There is a famous paradox that is called Collider Bias, consider that we have coding skills C that we can assume in a range of 1 to 5. Then we have another property which might be called cooperative skills that again is in a range of 1 to 5. We can classify people based on these skills, it's clear that two skills are independent. Now introduce a variable, the hiring status H that can be hired or not hired, and assume that Reply is going to hire you only if $C + S \geq 7$. We have several combinations to get the minimum requirement. We can come with a model in which we know that coding skill and collaborative skill influence the hiring state, if we plot the data knowing that we are hired we plot a correlation between C and S and we found a linear correlation and this correlation tells something that might not be true in the general population. This paradox comes up because we are conditioning on the collider.

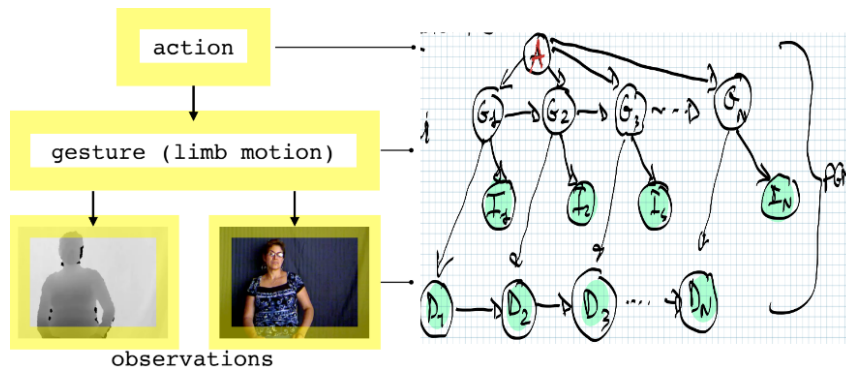


Chapter 3

Gestures

We can model gesture using PGM, we have several variables:

- Action
- Gesture that changes in time. A gesture depends on the gesture at time t-1
- Observation that also changes in time



We can adapt our Markov assumption and we can build a gesture level as kind of Markov problem in which G_2 only depends on G_1 and so on (head-to-tail structure). Obviously this is conditioned by A , given A (conditional probability) then I'll do the gesture.

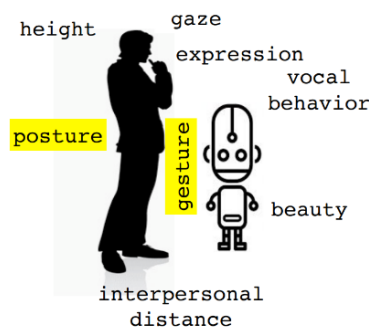
Then we have a complex generative process, because given the gesture (angle of gesture) we are generating the shadow, or more complicated the depth map or the video of gesture. That means that I might think that these two things (depth and video) are independent.

Remember that in this case our original problem was which action user is performing given the sequence of depth observations and video observations?

$$P(A|I_1, I_2, \dots, D_1, D_2)$$

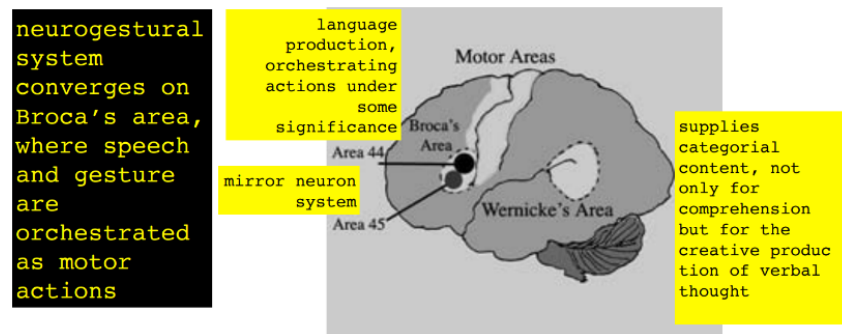
There are environments in which you can run this kind of PGM and use the model in inference time. This means that we are going down to up, from data to infer the action, we are not generating.

The problem is that gesture and posture can have very different meanings; they are para-linguistic communication.

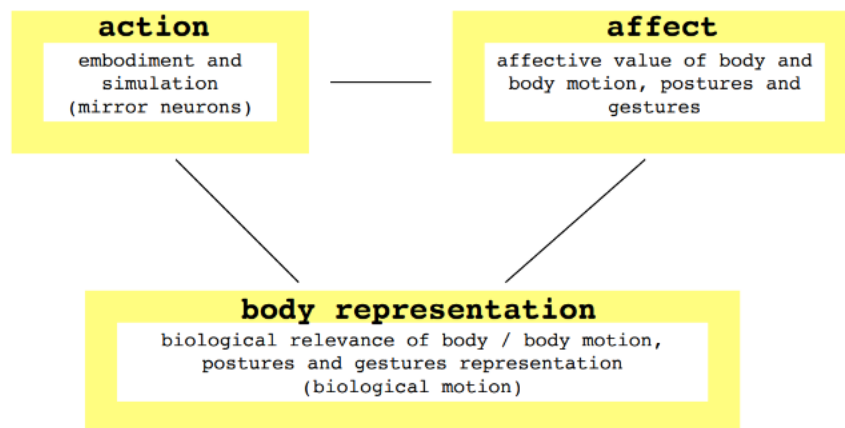


In gesture we might have some kind of linguistic signs, some culture specific (like ok) . Natural body gesture are much more involved in non verbal behaviour, are related to our affective state.

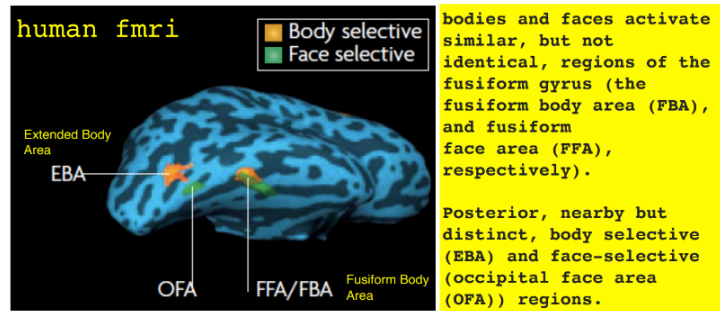
There is a clear evolutionary link between this kind of gesture and later on in the evolution what we acquired as language. What we find in the brain is that action are well represented, we have some kind of dedicated areas in our brain that are very efficient in detecting body movements. Then we have relevance, either in the representation.



How do I represent the action in my brain, the concept of action?



Body representation, we are very effective in recognize body, in our brains we have complicated relations between area involved certain actions/activity. We cannot say that there is an are that is specialized in process actions. But actually are ares that are more specialized than others. And also we have area that is more active when we see a face, or a body. These are really face and body selected areas, there are kind of are that we need because we are continuously interested in processing faces and body of other humans because is fundamental for our evolution.



We've seen the history of mirror neurons, we have this kind of neurons that are activated when we see someone execute actions. Actually we found either in premotor-areas, but there are also some in the parietal areas.

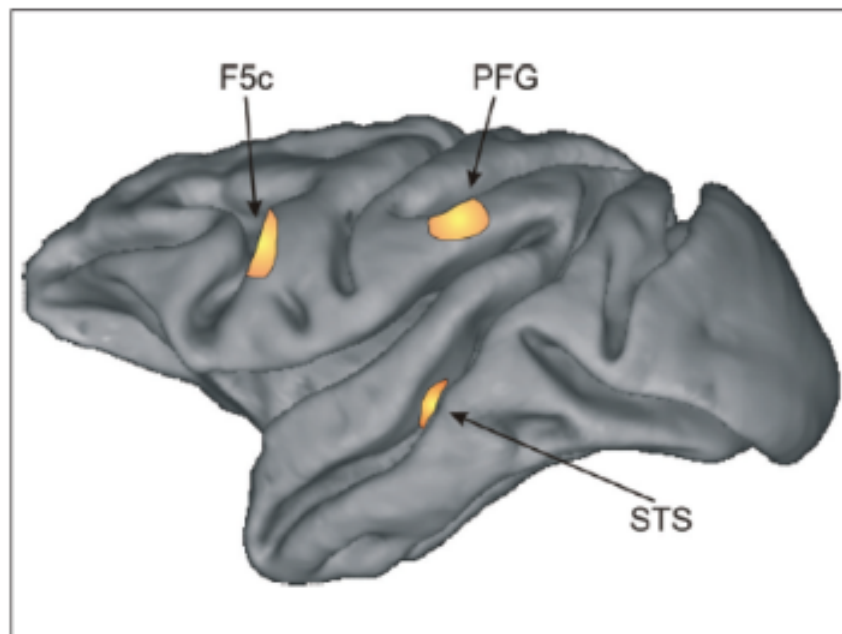
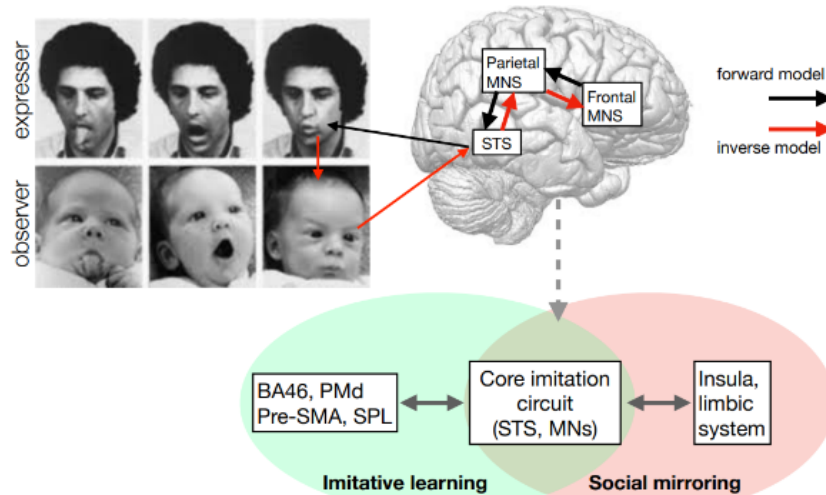
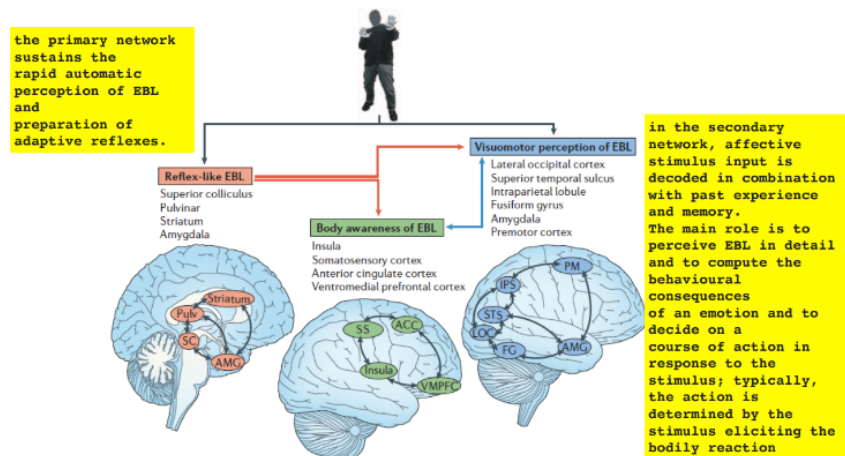


Figure 1. Areas in the monkey brain involved in action perception. The figure shows a lateral view of the left hemisphere of a macaque brain. The three highlighted areas contain neurons that discharge during observation of bodily movements. More specifically, neurons in the superior temporal sulcus (STS) exhibit purely visual responses and do not respond during the execution of movements. Neurons responding both during the observation and execution of goal-directed movements (i.e., mirror neurons) were found in areas F5c and PFG.

We can see the model of mirror neurons, and now we know that this works in a predictive way (generative). In this way, exactly like we seen in the PGM model we can go in both way, generating, confronting, inferring:

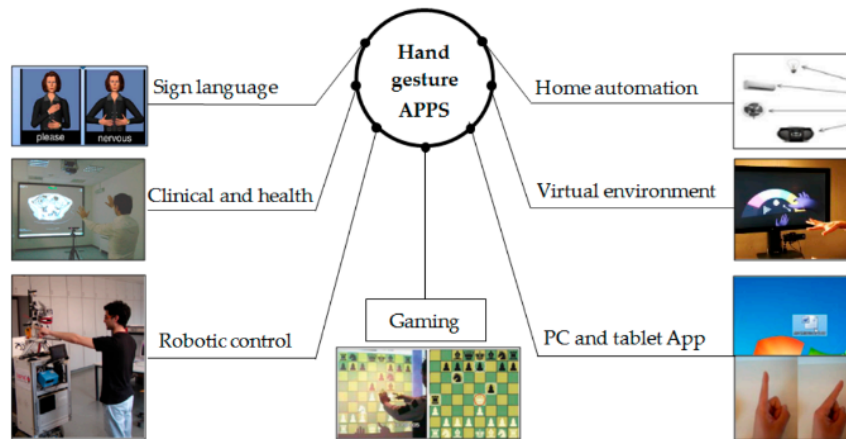


We can also express affect with our body, of course is more difficult:



3.1 Application

Given the generality of gesture we have variety of application. Processing gesture is not a solved problem but we now have a lot of tool that help us to recognize gesture and facial expression.



3.1.1 Model base

Build up a model and go using that model or use the model just a reference and go through neural network for recognition.

We can have kinematic model, using Kinect we have api to extract the feature. Or in other case we have to go through computer vision techniques that is more difficult because we can have occlusion problem and also segmentation could not be perfect.



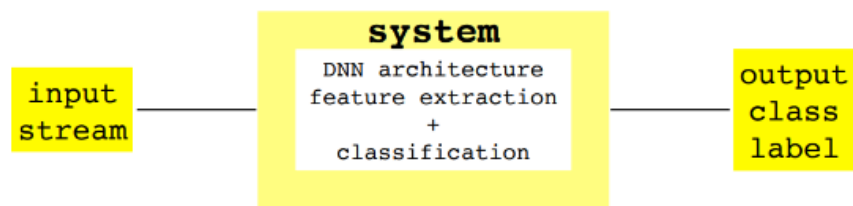
kinematic model



body parts

3.1.2 End-to-end

Are opposite of model base, we don't know how to model but we have a lot of data so we put the data in Neural Network and use these network to infer. Obviously to go through this solution we need to have a lot of data.



Chapter 4

Gaze and attention

We'll see a broad view of model that can compute attention. If we want model visual we want to take in account two basic question:

- Where/what people are looking, what attract their attention. That's mostly related to the content
- How do people look there? this is rarely see in the literature
- Why do people look there? this is based on internal motivation, something is more related to affective state while external reward is what is given from other people or the context. It's very difficult to address.

The essence of visual attention is based on what we are looking at is not a complete static scene but an ex-post reconstruction of our brain. Is a kind of continuous snapshot of the world.

We have an hypothesis, that is determined by some factors but they are so many that we have to distinguish from factor that we can model (something that we know, because for instance is part of the scene), but there are number of factors that are difficult to model (for instance motor actions), the sources that we cannot model act like noise. So the theory is: where we choose to look next at any given moment in time is not completely deterministic, but neither is it completely random.

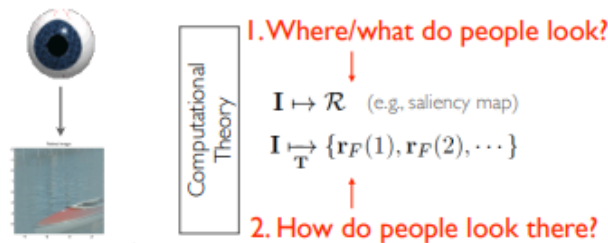
4.1 Models

Modelling means that we are going to have the sequence (the fixation sequence) of the position of our eyes in video. The model compute a sequence of coordinate.



The first model is the Itti and Koch model, this is the baseline for computational models. This model was, basically, the computational counter part of the Geladen model . We have an image and it's computing edge detection in different layers, they made a pyramidal structure in which they had a multi-resolution downsamplign the image in this way the captured the gray level contrast. They create 3 maps, color contrast, rotation and an other that i don't know. They collapse all the map in a master map. All this answer, in a certain way, where do people look (the white part of the master map).

The second part is how do people look, they had a sophisticated mechanism based on neural network, each neuron is an analog neuron, it's modeled by electrical circuit in which the capacity represent the level of saliency of the beginning, they integrate the capacity in time and when the integration reach a certain threshold they fire. The map provide the charge of the neurons, when the neuron is chosen and activate the neighbor discharging the area.

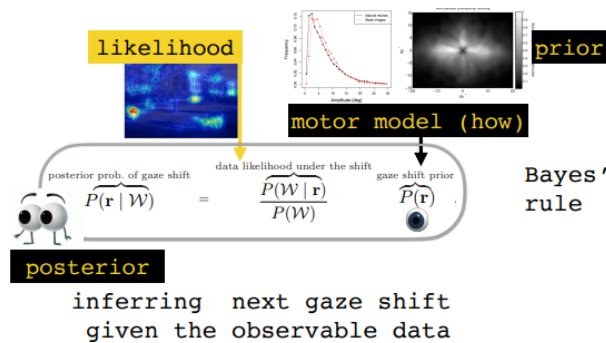


If we want to summarize we basically divide this problem in two steps:

- Compute some kind of representation of the image or video, saliency map
- Compute how the people look

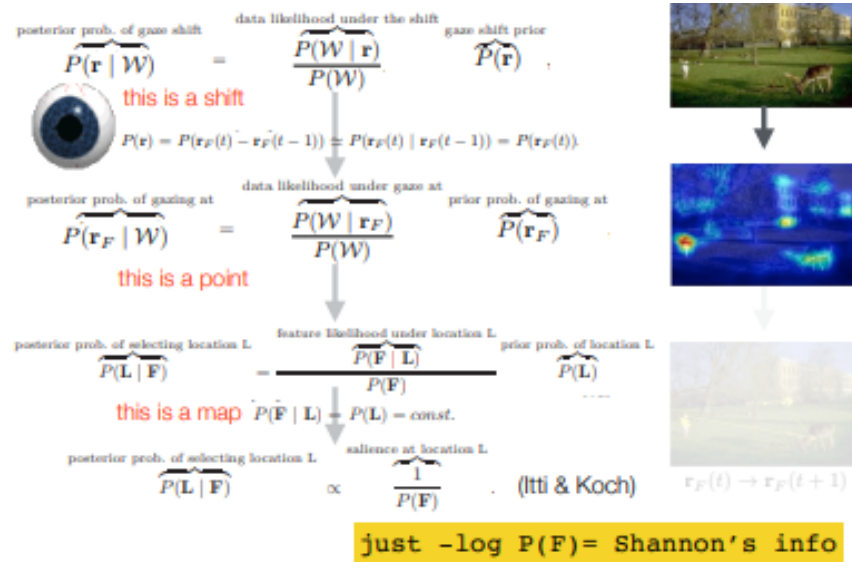
In video this bottom-up computation give better result because there is motion. In probabilistic term is the likelihood to have an interesting information given a certain location in the scene, what is the probability to observe an interesting feature given a location.

If we want to model the full stuff than we have to consider also the problem number 2, how we move? what we want to come out is a sequence of positions which is our gaze. The most general way of modeling both model is to consider Bayesian rule, given that I'm looking here and the property of the world what is the probability that the gaze shift from an other point. We have a prior on the shift, but what is that? if it's a movement the prior might consider the length, direction of the movement, it's called prior because some of these are actually a prior we know that we usually scan the ambient horizontally, rarely vertically and almost never diagonally. Also if we look at the way in which we look we know that there are some small movement that are more frequent than the longer, but longer movement can still occur.



Actually in literature we don't consider the shift but the probability to be in a point, if we don't consider the shift we cannot consider the shift, there is no movement in the second part. We can also remove the prior probability of location and consider only the

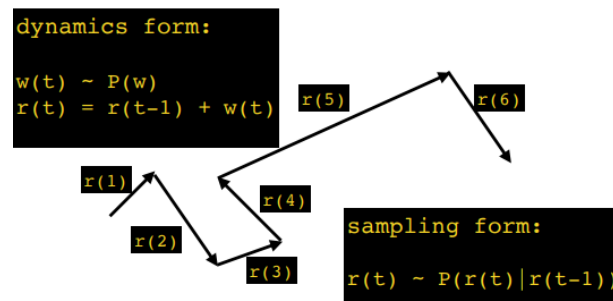
salience at location (the feature), this mean that if the probability of certain feature is low the silence is high and vice versa. This is called Shannon's information (the average is entropy)



Now the problem is that all of these, even if we removed the gaze. Come out a paper from Tatla and Vincent saying that there are biases (motor biases) that are very important for determine where we are looking at. They just used the motor biases of the users, not using the image, and completely blind just taking the prior the performance was exactly the same of the previous model.

4.2 Stochastic Process

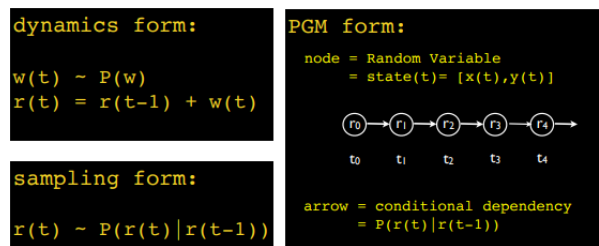
We sample a vector, we sample the probability of vector $\mathbf{r}$ at time t . When we sample the next point we are using Markov Assumption because we are considering only the previous point, the next position depends only on the previous point and the vector (that is what we are sampling). This stochastic process is similar of what happens whit our gaze. It's called a time series, is one realization of stochastic process.



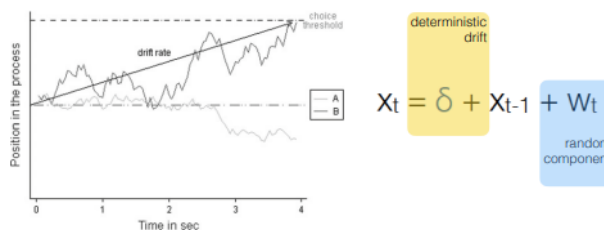
We might consider in two ways this modeling, from a pure probabilistic view we might consider this realization as the sampling of $\mathbf{r}_t$ from this conditional distribution, that is Markovian distribution, that tell how vector $\mathbf{r}$ at time t might be obtain given vector at time $t-1$. Then there is an equivalent way of seeing this generation of trajectory which is

more focused on dynamic form, given that I' in position t at time $t-1$ if I add the $w(t)$ I'm in the new position. The probabilistic part is the random vector that I add to the previous position, $w(t)$ is the noise that I sample to distribution. I'm in $t-1$ position, I sample random $w(t)$ and sum that vector to my position and move to next position. $P(w)$ is a Gaussian distribution.

An other possible representation is using the probabilistic graphical model we obtain a linear sequence head to tail which define a Markov Process:



We've seen also the drift diffusion model, if we use a drift we are also sampling random but we can in some sense put boundaries:



4.3 Markov Stochastic Process

If we don't use the Markov the future can taking to account all the previous model that is computational demanding.

We need to satisfy two assumption:

- The length of the jumps of the particles. And the distribution of the length is the same, this mean that particles are independent and distributions are identical. Each time we are sampling independently the Gaussian noise. They are IID.
- They must have finite moments. Moments are the mean, variance, all the things that we can compute integrating the distribution variables.

Given this two condition then we are within the condition of the central limit theory, this mean that we can sum the means and obtain a Gaussian distribution.

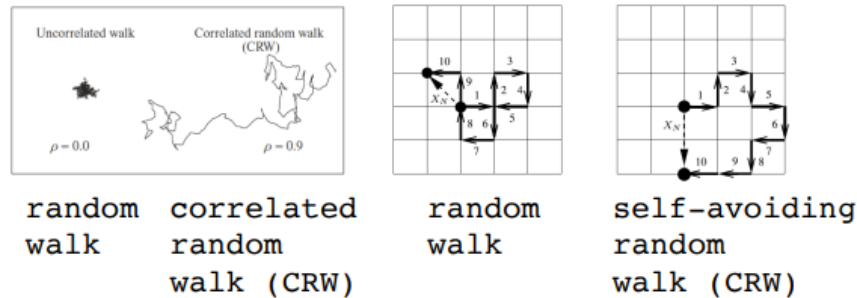
We are always working under the initial hypothesis that what we are considering is something which is not completely random.

We are now considering two case in which CLT hypostasis are violated:

- Non independent displacement: this generate a non Markov process. Particles are performing fractional Brownian motion
- non Gaussian process: this yield the Levys flies

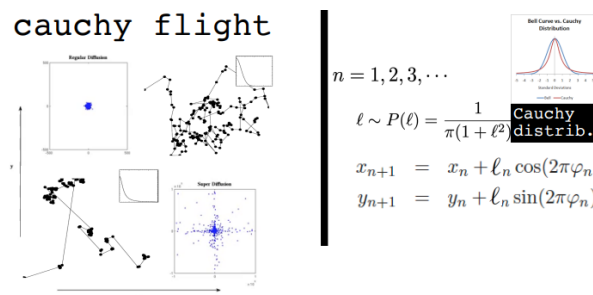
4.3.1 Non independent displacement

The typical example is the correlated random walk in which we have some constraints to the movement due to physic constraint in which we have correlation of sequence of position and if we measure the correlation we'll have higher correlation coefficient. In the example we see the example in which we cannot visit a position that we've already visited:



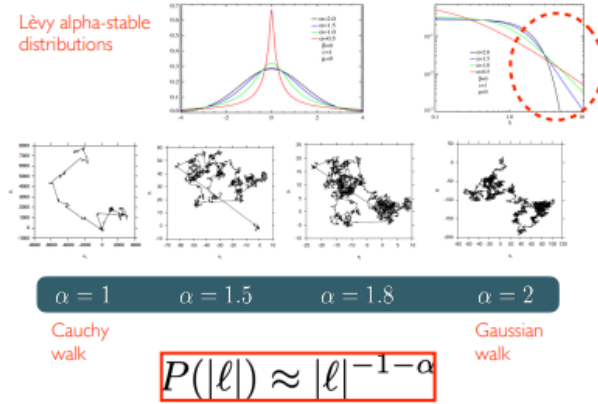
4.3.2 Non finite moments

we've seen a Cauchy flight in which we have local exploration and a long jump. We obtained this, we changed that the length, the noise that we are sampling to obtain the jump, is sampled by a Cauchy distribution that is part of Levy's distribution:



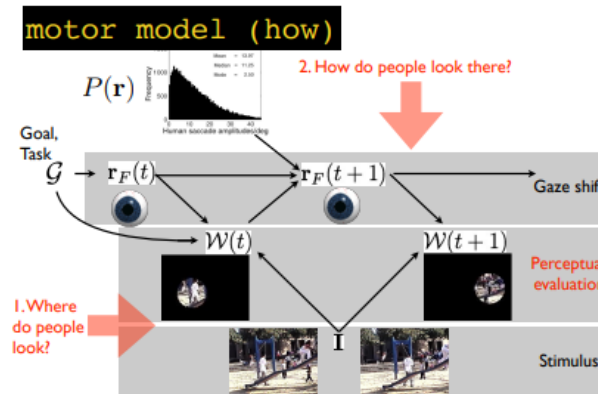
Using lot of particles using levy we obtain a super diffusion (very fast diffusion), that is used to describe very strange physical process.

We can model this distribution like that, it's not a close form solution so this is an approximation, note that if we inject $\alpha = 2$ we can obtain, more or less, the Gaussian distribution:



If we go back to the idea of motor biases we have this long tail distribution, that means that the long tail distribution are well suitable for describing the pattern that we have in movement.

Given that we have this formal statement of the pattern we can use it for modeling, note that we have constraint of the physical seeing and we know the prior of the lenght, we have can use it as a prior:



Next step is to extend this idea to the videos, in video we have an other characteristic we do a smooth pursue movement basically we are moving our eyes looking a moving thing. The principle is the same, we have the basic equation, the deterministic components that is the distance between attractor that acts like a sort of gravitational forces clearly the nearest they are the more are attractive, then we have the stochastic component here in order to taking into account all the patterns what they allowed for at least 3 alpha stable distribution which they are related to each movement.

4.4 Foraging

Studies showing different behavior of different animals due to level of dopamine. Most interesting is that this basic behavior are related to long evolution and transform in the sense that what was once true foraging behavior also shape also cognitive behavior.

The idea of foraging is very nice, it's a well studied subject:

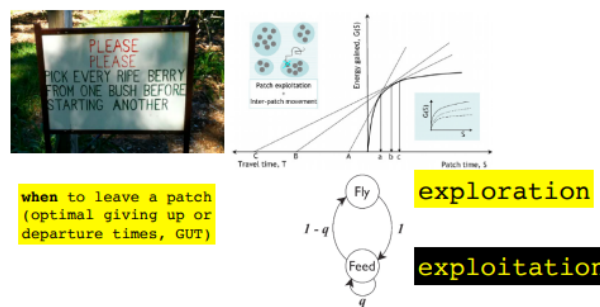
- What prey to take: optimal diet choice
- What patch type to search: optimal patch choice
- When to leave a patch: optimal giving up to departure times, GUT. We have theorems to model this issue
- How to move between patches: optimal movements

These are exactly the same problems in gaze theory. We have Charnov's that it says that is always a trade off, the curve is basically the energy of the animals (eye + brain) once enter in the patch starts gain energy. The travel time is a important variable, because the longer the distance more likely the animal will stay within the patch (because changing patch will consume energy).

We might summarize this foraging behavior in term of two general features:

- Exploration: moving to an other patch
- Exploitation: correspond to the within patch

Always the behavior is a trade of between these two variables.



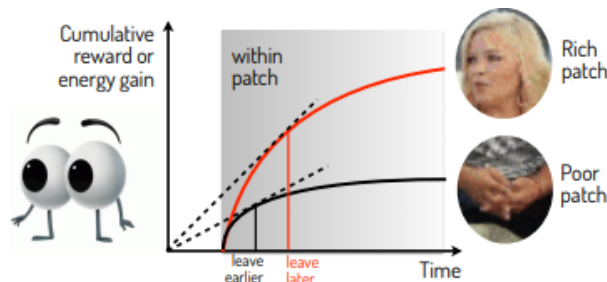
Now in our cycle in which we have eye movement we can embed all this knowledge. They studied the problem of what happens when we are looking some kind of scene in which there is a conversation. Of course what happens is that we try to focus on the speaking person. From a computational point of view what happens is that the input of the model was a multi-modal stream (video and audio). At the beginning you need to have some patches, the condition to apply the foraging approach, in these case the patches are not only visual but also multi-modal. That means that there is a lot of pre-processing but we'll skip this part. Also here we are in a problem in which we have temporal dynamics, these patches are better not to be detected analyzing the video frame each frame because what happens at time t depends on what happens on time $t-1$, and also we might have some change of attention importance in respect of each patch depending on habituation time. Then once you have this setting, then you might have some techniques in order to infer the current goal and value, you would like to infer from the data the preferences of the observer. Once you have all this information you are left to the foraging problem, how would you choose a patch, how long do you stay and how do you move, the gaze problem.

Actually what you might observe when you look at the data you might see that in these scene, for instance low level silencie play a very limited role. What happens is that the model concentrate on these patches:

- Speakers
- non speaker
- gesture

This because these patches have a social value.

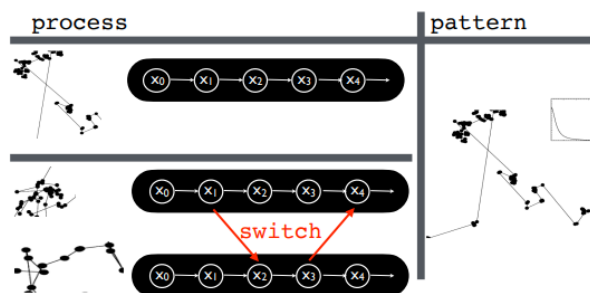
Once you have that you can apply and formalize the information reachness within each patch, and what we see that the reach patch (with high social value) will be abandoned later with respect to patches that have lower social value.



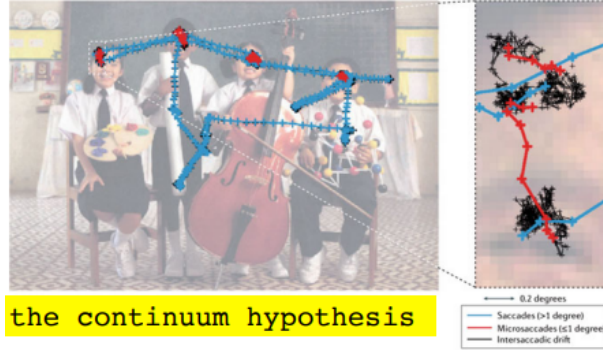
4.4.1 Composite process

In the model we've seen our main effort was to use statistical model was trying to explain the pattern we are observing that we detected with eye tracker, this is a phenomenological model because we are not taking in account the reason why this happens (short movements and then long jump). Some school say that this is not only a pattern but also a process, another school say that these the pattern but the process to arrive to that pattern need not to be as the pattern, they say that can obtain this pattern composing two other pattern, rather than having a unique model here we have a composite double stochastic process.

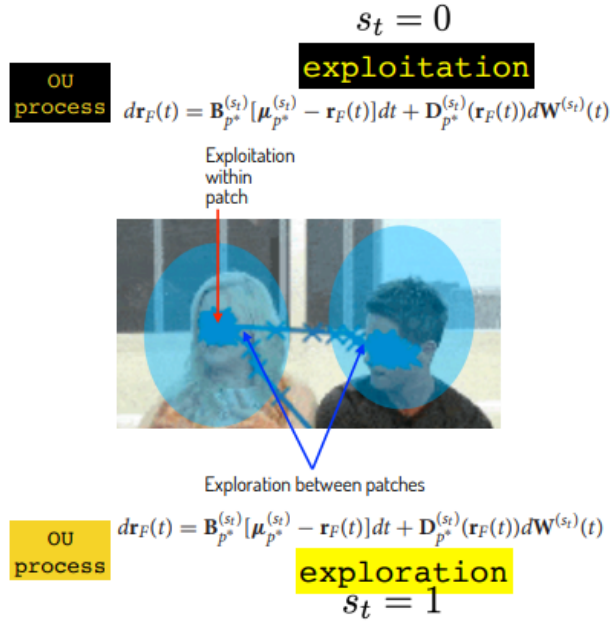
These two simple process are two braunian motion (wieneer process), we have a braunian walk that work locally (small steps) and then we can switch in a flight mode in which we have a large braunian motion, it's not a Levy process.



The advantage to the composite process is that in all the states we are working with a Gaussian distribution. This is not only a matter of mathematical convenience but there is a deep reason, the continuum hypothesis, it's like we have a two scale both with braunian motion:



In the experiment of attention on the speaker they have two state, the two model in composition model. Both motion, the small fixtional movement and the large, can be described with the same equation but different parameters. In the equation there is a deterministic part and a stochastic part, to decide the next position is easy, each face in the video is an attractor, then there is a stochastic part that is a Gaussian sampling that is conditioned by a diffusion parameter and obv the covariance of the matrix (larger covariance means larger jumping of the motion), when I go to the exploitation we have the exactly the same equation but different parameter, the attraction will be the items (eyes, mouth...) that is driven by the distance of the attraction and we again have the Gaussian motion but the local diffusion and also the Gaussian will be smaller. A switching parameter allow to inter-live the two models, according to some condition we'll refer to one or other stochastic equation.



An interesting result of this model is that the model doesn't model the duration but it's really similar to the real duration.

How can we use these models? we might use the model for practical purposes, we do some experiment and do some measurements to characterize attention of different subject in medical environment, or we might eye track people during a commercial spot for marketing. The first study is composed as a given picture, infer the parameters of fixation model and the saccade model.

The second example was more complex example, they want to predict when the people are anxious or not given some conditions. In this experiment they give some stimula.

4.5 Leveraging modern AI tools

You have some verbal description of what you know in the world, now we have a model that use deep neural network to produce a description of an image, what we can do is to invert the process, we can give the description and the model will generate the image.

All these are based on language model, the idea is that they are not Markov model but they use transforms the predict the next word using all the previous word, full auto-regressive model.

These transformers have been trained not only on the text but also for video, we might tokenize also images. This is a very nice because they can perform a text base saliency map.

Chapter 5

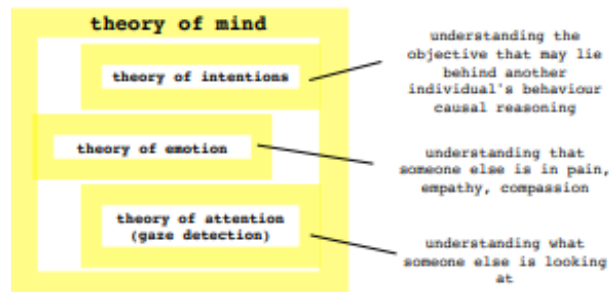
Gaze for social signal

We use gaze as a social signal, and also as interpersonal communication.

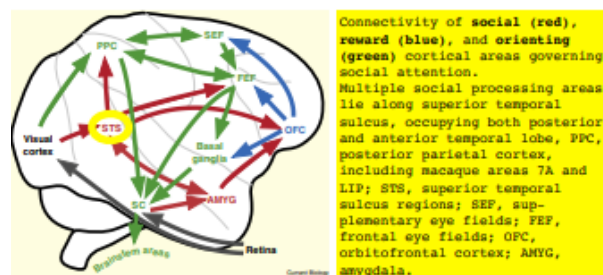
The basic idea is that in this case gaze is a sort of communication channel, there is huge literature on this. Gaze is used as affiliative cue, an important point is that we have even further develop this kind of capabilities because in some way we are free of some problems that you might have in gazing individual for instance in apes societies in which there is strict rule of society.

5.1 Theory of mind

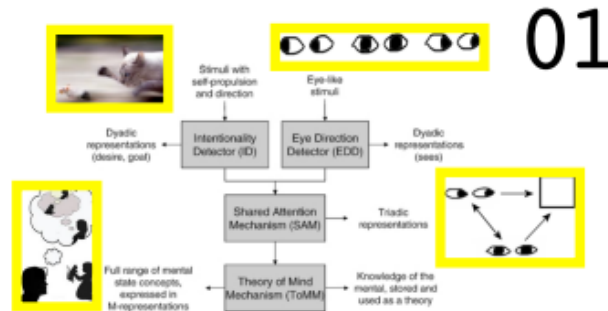
If the theory of mind is our way to infer that other people reasoning for full theory of mind we need language because it fully exploit the recursive property that language offers. But there are building blocks in TOE and theory of attention and theory of emotion deal with gaze.



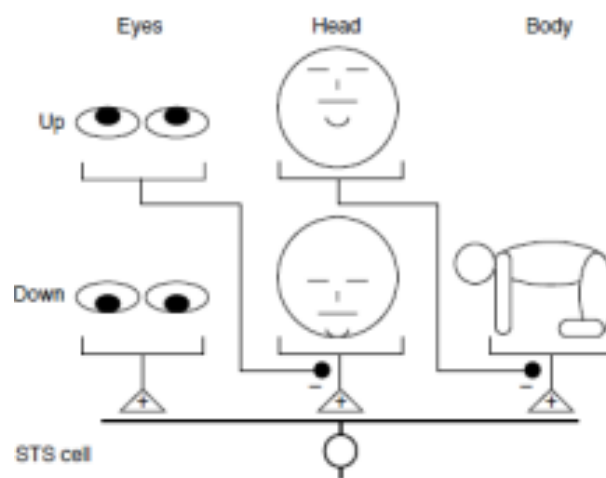
This scheme is also true at neurological level, in this scheme we have tree paths:



We have a model that some way explain how the gaze works, in first we have intentionality detector (I understand what you want) and eye direction detector (rough evaluation of where other people are looking). The next step is shared attention (me and you are looking there, we are joining our attention mechanism, this is a very basic step of communicating with others). The last step is generalizing the TOM, as human we have also language.



We have an other model, the Parret & Emery model that contradict the previous eye detection model providing a combination of other part, like the body or head direction:



If we are involved in a project like this we don't have to invent anything from scratch but we have some instruments like OpenFace for gaze tracking. It performs eye detection, facial landmark analysis and gaze detection using the combination of head direction and eyes direction.

Chapter 6

Machine Learning

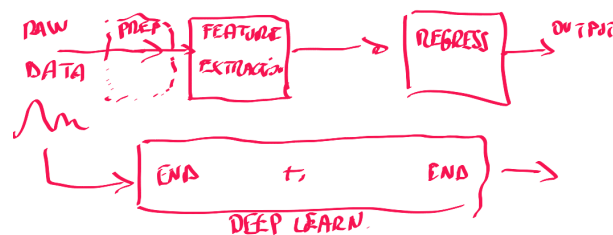
Machine Learning is a part of the AI and recently has been characterized with deep learning. The idea of this section is to understand the basic problems of machine learning, this will help us in our career to learning new things having a solid background.

In this field we always start from data, they are fundamental. We might have a certain number of data, or vector, and this is usually called dataset (in this notation the underscore means a vector):

$$D = \{x_n\}_{n=1}^N$$

Since machine learning means learning the parameters of the model, what we'll do is usually divide this dataset in a train and test and there are many ways of doing this. A well known problem is the bias, always in the way data are collect we have a bias in dataset because the data are a representation of the same distribution (the same population). There is big rule, trash in trash out, if we don't provide good features we'll obtain poor results.

In deep learning way usually we have this pipeline, row data (with some pre processing) are provided to a black box that compute the feature extraction. Then these feature go in the regression model that give us the result:



The problem is that even if we give a huge amount of data this end to end architecture will have million or parameters, but if we use a lot of parameters we are likening to over-fit the parameter, we are extremely adapting the model to these data. This means that when we change the dataset (from training to completely new) the performance of the model will decrease, we know that with deep learning techniques this effect is moderated.

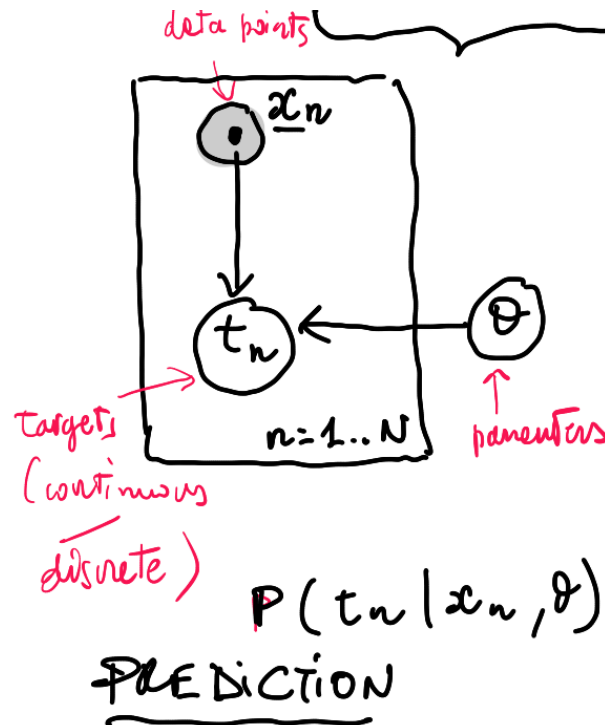
If we want to use a taxonomy for machine learning we might have cluster of methods:

- Continuous output
- Discrete output: might be classification
- Supervised setting: we might have data and labels for each of data $\{x_n, l_n\}$
- Unsupervised setting

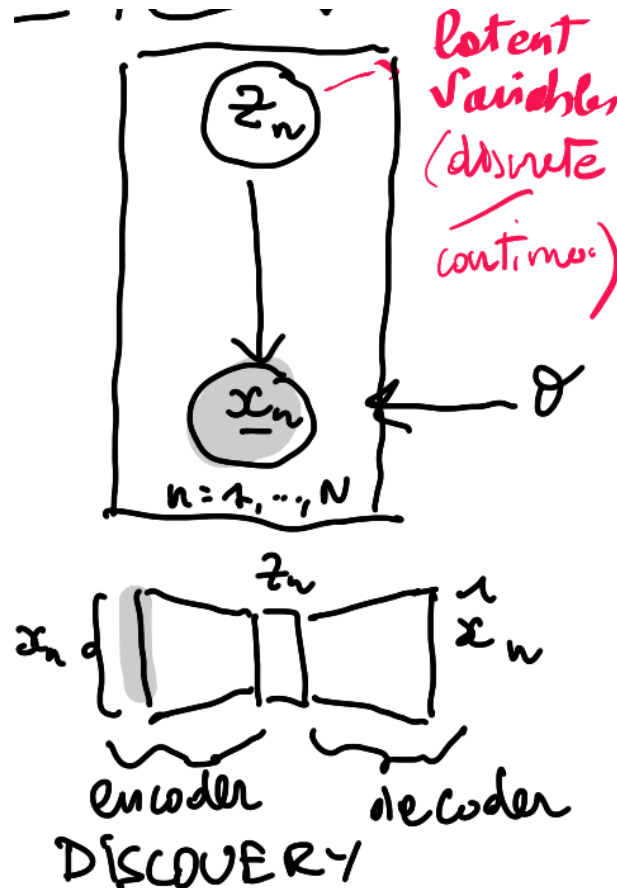
	Supervised	Unsupervised
Continuous Output	REGRESSION	DIMENSIONALITY REDUCTION
Discrete Output	CLASSIFICATION	CLUSTERING

If we want to consider the supervised part of the scheme we might draw a PGM in which we have a given data point (it's not a random variable), what we have to do is from this depending on what are the feature, is would like to have an output (continuous or discrete) that is a random variable also that is single point dependent. Here there is a

plate notation, that is basically a for loop, we are repeating from 1 to n. And we have another random variable that is the parameters of the model. This basic model capture all the algorithms, the target could be continuous or discrete but the supervised model has the same structure. The problem, again, is to predict t_n given x_n and θ $P(t_n|x_n, \theta)$. In fact in many textbook this is called prediction model.



What about the unsupervised model? we might have a similar structure, the variable that we want to infer is called latent variable, and this generate some data under some parameters set. Usually here we have available only the data. It's slightly different and more complex problem. Given the data what we want to do is given the data infer the probability of this latent variable, that is a variable that we are not observing. In case of continuous variable is the variable that describe the shape of the shadow in 2 dimension (variable in PCS) this is the encoder (or auto-encoder part) we want to map data to a low dimensionality space. Once we have this encoder part (map the data in a lower dimensionality space, like learn how to map a face with a low representation) we can use this z_n as a generative model to produce x_n . The latent variable might be discrete or continuous, this problem is called discover problem., because we don't have the label and we have to discover the label.



There is a third model, the decision making, that contains the reinforcement learning algorithm and more recently the problem of causality. The central problem is the inference problem, because from a Bayesian stand point there is no difference between learning and infer a parameter.

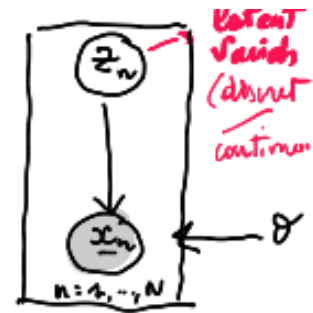
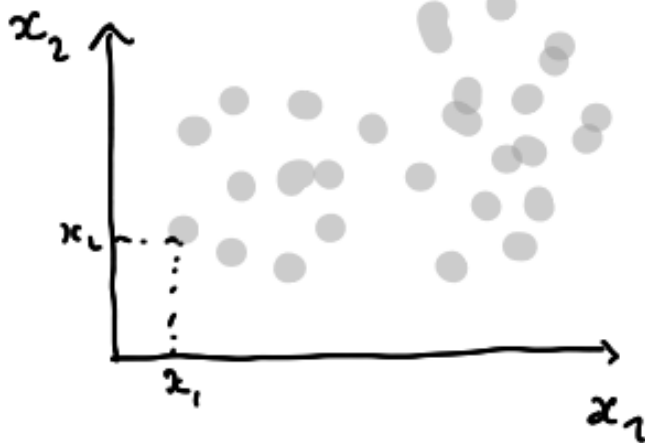
Let's see the clustering example. Clustering would be this problem here, for instance we might assume that we have 2 dimension (spacial clustering) and our data might be a numbers of point in this space. We only have the data not the label. We also can assume a model, in our case that there are 2 groups in this data this is an hard problem but there are methods to select this model (heuristics) or we can have hypothesis. The clustering problem amounts to the fact that what we want to do is aggregate and we might expect that our algorithm in the end do some kind of classification, basically you are performing the classification between 2 groups without having the label. We are assigning a latent variable to each of the point, this is a difficult problem. A famous algorithm is k-means, that is a deterministic solution of this problem in which we have the two cluster and the mean of each cluster that at the beginning we don't know and we put them causally and color the nearest points, after that we recompute the mean and recolor. It's a simplification of expectation and determination algorithms. Under the assumption that this data have a Gaussian distribution and mean that we compute is the mean of the Gaussian. From the probabilistic point of view we choose which Gaussian we want to use to sample and once we have chosen we generate the colors given the x_n that

we sampled:

$$z_n \sim P(z_n)$$

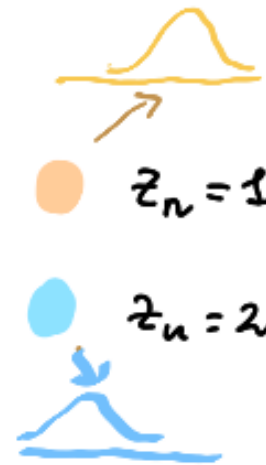
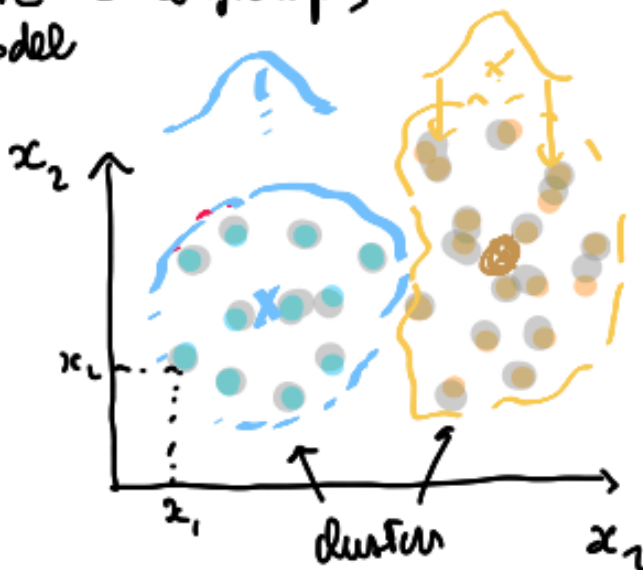
$$x_n \sim P(x_n|z)$$

Example : clustering



$$\underline{x}_n = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$M = 2$ groups
model



$$\begin{cases} \textcircled{1} z_n \sim P(z_n) \\ \textcircled{2} x_n \sim P(x_n|z_n) \end{cases}$$

a generative view of clustering

6.1 Regression

In regression tasks, we start with a dataset comprising points, each associated with a continuous target label. The objective is to build a model that correctly maps each point to its label. In linear regression, the estimated label is given by:

$$t_n = w_0 + w_1 x_n + \epsilon_n$$

where ϵ_n represents noise in the measurements, and w_0, w_1 are the model parameters to be learned during training. The model is linear with respect to the parameters (not the data); different parameter values can yield various line shapes.

For deterministic models, noise is not considered. In such cases, for each point:

$$t_n = w_0 + w_1 x_n = [w_0 \ w_1]^T [1 \ x_n] = \underline{w}^T \underline{x}_n$$

Here, t_n , is a scalar. Collectively, the formulation becomes:

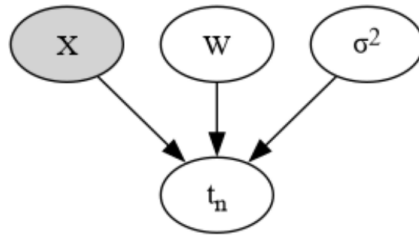
$$\mathbf{t} = \mathbf{X} \mathbf{w}$$

where $\mathbf{t}$ is an $N \cdot 1$ vector, $\mathbf{X}$ is an $N \cdot 2$ matrix, and $\mathbf{w}$ is a $2 \cdot 1$ vector. Estimating $\mathbf{w}$ directly by inverting $\mathbf{X}$ is generally impractical because $\mathbf{X}$ may not be invertible.

6.2 Probabilistic generative models

Probabilistic generative models provide an alternative solution and are stochastic by nature. Deterministic models neglect measurement noise σ^2 , whereas probabilistic models incorporate it to quantify prediction uncertainty.

Given data, we seek the probability of a label t_n given parameters $\mathbf{w}$ and noise σ^2 :



$$P(t_n | \underline{x}_n, \underline{w}, \sigma^2)$$

Maximum Likelihood Estimation (MLE)

Using MLE:

$$P(\mathbf{t} | \mathbf{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^N P(t_n | \mathbf{x}_n, \mathbf{w}, \sigma^2)$$

which forms the likelihood function $L(\mathbf{w}, \sigma^2)$. Denoting $\theta = \{\mathbf{w}, \sigma^2\}$, we solve:

$$\hat{\theta} = \arg \max_{\theta} L(\theta)$$

If $L(\theta)$ is differentiable, we find $\hat{\theta}$ by solving:

$$\frac{\partial L(\theta)}{\partial \theta} = 0$$

Noise Model

For the noise model, we often use a Gaussian distribution:

$$\epsilon_n \sim \mathcal{N}(0, \sigma^2)$$

Thus:

$$t_n \sim \mathcal{N}(t_n \mid \mathbf{w}^T \mathbf{x}_n, \sigma^2)$$

Combining MLE, the likelihood becomes:

$$L(\mathbf{w}, \sigma^2) = \prod_{n=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(t_n - \mathbf{w}^T \mathbf{x}_n)^2\right)$$

Taking the log-likelihood:

$$\ln L(\mathbf{w}, \sigma^2) = -\frac{N}{2} \ln 2\pi - \frac{N}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{n=1}^N (t_n - \mathbf{w}^T \mathbf{x}_n)^2$$

The term $(t_n - \mathbf{w}^T \mathbf{x}_n)^2$ is the squared error. Thus, minimizing the squared error corresponds to MLE with a Gaussian noise model:

$$\sum_{n=1}^N (t_n - \mathbf{w}^T \mathbf{x}_n)^2 = (\mathbf{t} - \mathbf{X}\mathbf{w})^T (\mathbf{t} - \mathbf{X}\mathbf{w})$$

Defining $(\mathbf{t} - \mathbf{X}\mathbf{w})$ as $\mathbf{e}(\mathbf{w})$, the log-likelihood simplifies to:

$$\ln L(\mathbf{w}, \sigma^2) \propto \|\mathbf{e}(\mathbf{w})\|^2$$

Summary

- Linear models are expressed in terms of parameters; data points are generated using the model.
- Deterministic models do not consider noise in measurements.
- Stochastic models incorporate a probabilistic error model.
- With a Gaussian noise model, points are distributed around a line, and MLE provides a squared error form.
- Parameters are estimated as scalars rather than random variables (as in Bayesian models).

Parameter estimate, OLS

Estimating the model, the line in this case, that means estimate the vector of parameters. This can be done classically taking the derivative of the function in respect to the parameters:

$$\begin{aligned}\frac{\partial l}{\partial w} &= \frac{\partial}{\partial w} \left(-\frac{N}{2} \ln 2\pi - \frac{N}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} (\mathbf{t} - \mathbf{X}\mathbf{w})^T (\mathbf{t} - \mathbf{X}\mathbf{w}) \right) = 0 \\ \frac{\partial}{\partial w} \left(-\frac{1}{2\sigma^2} (t - Xw)^T (t - Xw) \right) &= 0 \\ \frac{1}{2\sigma^2} \frac{\partial}{\partial w} (-J(w)) &= 0 \\ \hat{w} &= \operatorname{argmax} L(w, \sigma^2) \\ &= \operatorname{argmin} J(w)\end{aligned}$$

We can manipulate our min error function:

$$\begin{aligned}J(w)e^T e &= (t - xw)^T (t - xw) \\ &= (t^T - w^T x^T)(t - xw) \\ &= t^T t - t^T Xw - w^T x^T t + w^T x^T xw \\ &= t^T t - 2w^T x^T t + w^T x^T w \\ &= t^T t - 2w^T x^T t + w^T Cw \\ \frac{\partial F(w)}{\partial w} &= \frac{\partial}{\partial w} (t^T t - 2w^T x^T t + w^T Cw) \\ &= -2 \frac{\partial}{\partial w} w^T x^T t + \frac{\partial}{\partial w} w^T Cw \\ &= -2x^T t + 2Cw = 0 \\ &\rightarrow x^T t = Cw \\ x^T t &= x^T xw \\ (x^T x)^{-1} x^T t &= (x^T x)^{-1} (x^T x)w \\ \hat{w} &= (x^T x)^{-1} x^T t\end{aligned}$$

t are the labels, the $x^T x$ is the covariance matrix that we have, so what we are doing is given the data predicting the parameters. This is the classical mean square solution to the regression problem.

To solve this equation we used this cheat sheet that solve derivative of matrix:

$F(w)$	$\frac{\partial}{\partial w} F(w)$
$w^T x$	x
$x^T w$	x
$w^T w$	$2w$
$w^T Cw$	$2Cw$

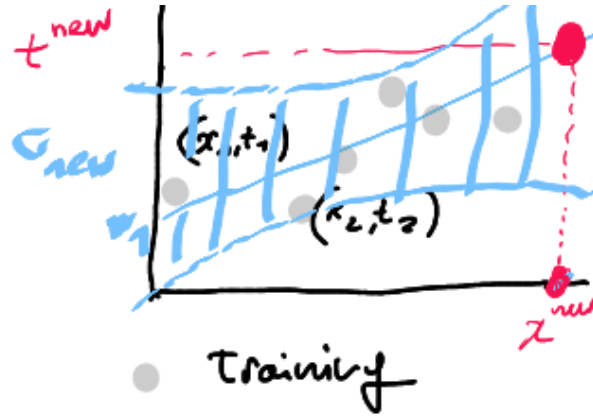
Now we must solve a second problem, estimating σ^2 given the estimated parameters $\hat{w} = (x^T x)^{-1} x^T t$:

$$\begin{aligned}
\frac{\partial l}{\partial \sigma^2} &= \frac{\partial}{\partial \sigma^2} \left(-\frac{N}{2} \ln 2\pi - \frac{N}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} (t - xw)^T (t - xw) \right) \\
&= -\frac{N}{2} \frac{1}{\sigma^2} + \frac{1}{2\sigma^4} (t - xw)^T (t - xw) = 0 \\
\rightarrow \hat{\sigma}^2 &= \frac{(t - x\hat{w})(t - x\hat{w})^T}{N} \\
&= \frac{(t - x(x^T x)^{-1} x^T t)^T (t - x(x^T x)^{-1} x^T t)}{N} \\
\hat{\sigma}^2 &= \frac{1}{N} (t^T t - t^T \hat{t})
\end{aligned}$$

$t^T t$ is cov true labels $t^T \hat{t}$ cov between true and predicted

Once we have the OLS classical (predicted parameters $\hat{w}$) and the estimation of the probability model (σ^2) we can do the prediction:

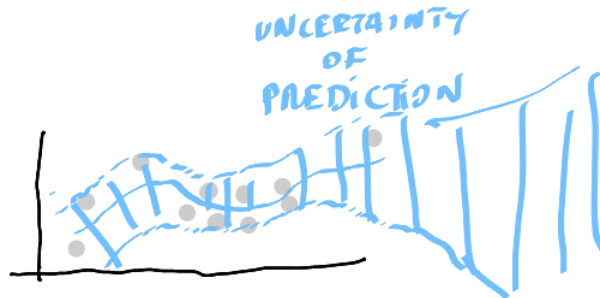
$$\hat{t}_{NEW} = \hat{w} x_{NEW}$$



We can also predict the standard deviation of the predicted values:

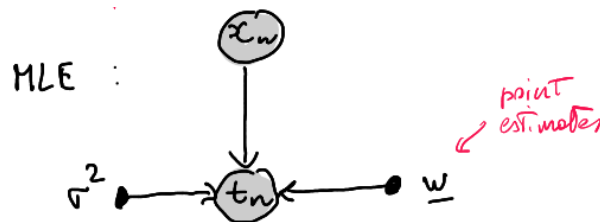
$$\text{var}(t_{NEW}) = x_{NEW}^T \text{cov}(\hat{w}) x_{NEW} = x_{NEW}^T \hat{\sigma}^2 (x^T x)^{-1} x_{NEW}$$

The big difference in respect to deterministic model here we have a close form solution for the estimate of uncertainty of prediction $\hat{\sigma}^2 (x^T x)^{-1}$. The shape in the graph tells us how precise is the model to predict new points.

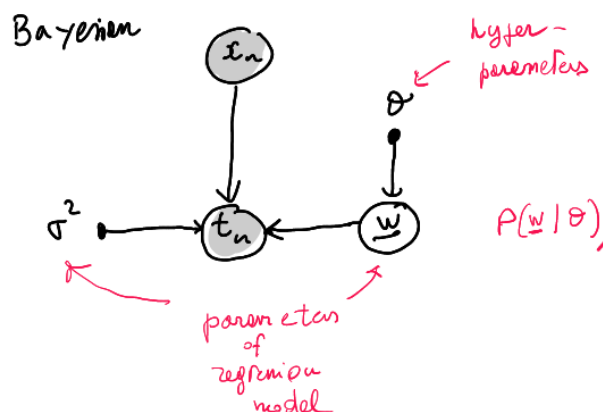


Bayesian regression

So the model that we want is in this form $P(t|x, w, \sigma^2)$. In MLE we have observed variables with labels and then we have point estimated from w and σ^2 that are point and not node because these are point estimates:



The Bayesian approach is more general, because again we have our data (x_n) and label (t_n) that are observed. Again we have the σ^2 but most important is that now the weight are random variable, this means that there are probability distribution that have some parameters that to keep things simple we identify as points (σ), these are called hyper-parameters:

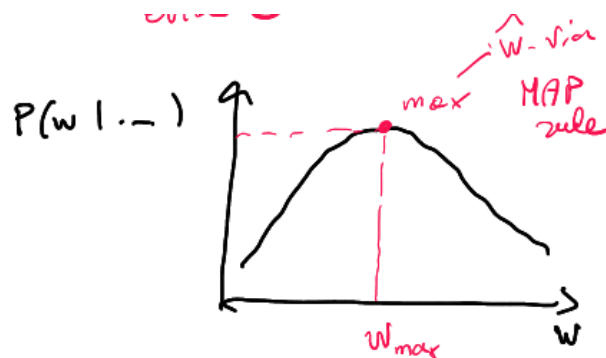


In the Bayesian setting the parameters learning, which will accomplish to MLE, is equal to the inference of parameters. In the general case our parameters inference is actually the inference (calculation of the posterior over the parameters), which will determine this rule:

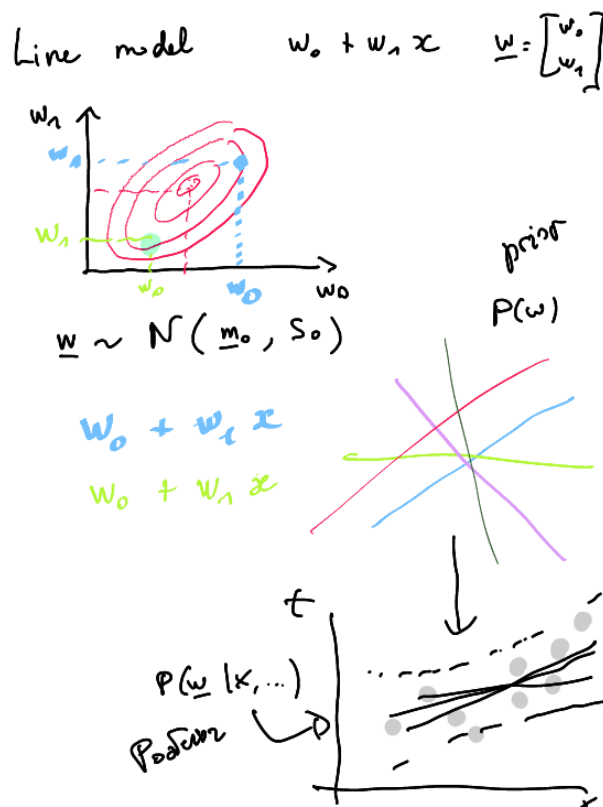
$$P(w|t, x, \sigma^2) = \frac{P(t|x, w, \sigma^2)P(w|\sigma)}{P(t|x)}$$

Where $P(t|x, w, \sigma^2)$ is the likelihood function and $P(w|\sigma)$ is the prior distribution over the parameters (hyper-parameters) and $P(t|x)$ is the evidence. So what we compute is the posterior distribution over the parameters $P(t|x, w, \sigma^2)$.

If we take the max of the distribution that would give rise as the estimate via Maximum a Posteriori rule:



What does mean if I sample a point there? it's different having a distribution of parameters, we sampled one number from a Gaussian distribution, this number means that we are sampling w_0 and w_1 , that means that we are sampling a line. Assume that we are sampling an other point (green), we'll have other w_0 and w_1 that correspond to an other line. So each time I sample a point what I'm actually doing is sampling a line (possible model). What happens here is that we are sampling without context because it's called the prior model.



When we fit Bayesian model actually we are not getting one line (like we do in MLE), is like we have a family of possible lines (not just one) and all of these are compatible with the data. The uncertainty is provided by the covariance of the posterior distribution.

6.2.1 MAP (Maximum a posteriori)

We are seeing the solution for Bayesian Regression only considering the maximum point of the distribution.

$$\begin{aligned}\hat{\underline{w}}_{map} &= \arg \max [P(\underline{w}|\underline{t}, x, \sigma^2)] \\ &\simeq \arg \max P(\underline{t}|x, \underline{w}, \tau^2) P(w|\sigma)\end{aligned}$$

The form of the likelihood is the same as the one we used in maximum likelihood. In order to set the prior we need to define the parameter theta. In order to do that we'll assume a Gaussian model on the w (parameters), that means that theta will be the mean and the covariance matrix, that means that our $P(w|\theta)$ will be equal to a normal distribution:

$$P(w|\theta) = N(\underline{w}|\underline{m}_0, S_0)$$

$$S_0 =$$

So this will be our model or prior distribution. Note that we found the identity $\text{map}(\text{matrix}) \mathbb{1}$ in this model. the exact form of our function will be the same as before:

$$\arg \max \left[\sum_{n=1}^N \ln N(t_n | \underline{w}^T \underline{x}_n, \tau^2) + \sum_{k=1}^k \ln N(\underline{w}_k | 0, \tau_k) \right]$$

Now we have already the development to this function so we don't need to recompute everything because the first part is exactly what we have in the likelihood case:

$$\begin{aligned}\arg \max & \left[-\frac{1}{2\tau^2} (\underline{t} - x\underline{w})^T (\underline{t} - x\underline{w}) - \frac{1}{2\tau^2} \underline{w}^T \underline{w} \right] \\ \implies & \frac{\partial}{\partial \underline{w}} \left[-\frac{1}{2\tau^2} (\underline{t} - x\underline{w})^T (\underline{t} - x\underline{w}) \right] + -\frac{1}{2\tau^2} \frac{\partial}{\partial \underline{w}} \underline{w}^T \underline{w} = 0\end{aligned}$$

The first derivative we've already computed because it's the OLS, and second derivative is equal to w^T . So what we have in the end:

$$\frac{1}{\sigma^2} (\underline{w}^T x^T x - \underline{t}^T x) + \frac{1}{\tau^2} w^T = 0$$

So it's exactly like the maximum likelihood except the term that is related to the prior distribution. Now we arrange a bit this expression, we factorize in respect to the w^T :

$$\begin{aligned}\implies & \underline{w}^T \left(\frac{1}{\sigma^2} x^T x + \frac{1}{\tau^2} \mathbb{1} \right) - \frac{1}{\sigma^2} \underline{t}^T x = 0 \\ \implies & \underline{w}^T \left(x^T x + \frac{\sigma^2}{\tau^2} \mathbb{1} \right) = \underline{t}^T x \\ \implies & w^T = \underline{t}^T x \left(x^T x + \frac{\sigma^2}{\tau^2} \mathbb{1} \right)^{-1} \\ \implies & \underline{w} = \left(x^T x + \frac{\sigma^2}{\tau^2} \mathbb{1} \right)^{-1} x^T \underline{t}\end{aligned}$$

Interlude - Regularization

In the classic regression (OLS) what we do is to minimize the error function:

$$J(\underline{w}) = ||\underline{t} - x\underline{w}||^2$$

In this case we have the problem of model selection, we might consider a general form for our data which is a polynomial form:

$$\begin{aligned} f(\underline{x} : \underline{w}) &= \sum_{k=1}^k w_k x^k \\ k=1 \quad f(x, \underline{w}) &= w_0 + w_1 x^1 \\ k=2 \quad f(x, \underline{w}) &= w_0 + w_1 x + w_2 x^2 \\ k=3 \quad f(x, \underline{w}) &= w_0 + w_1 x + w_2 x^2 + w_3 x^3 \end{aligned}$$

We notice that when we move to a model with a lot of parameters we find to have a model that goes all over the point so zero error, but this is a test set. This problem is called overfitting and that it's what we should try to avoid.

So what people did was to introduce the concept of regularization, amount to other term taking in account the problem in which OLS is augmented with regularization term:

$$J(\underline{w}) = ||\underline{t} - x\underline{w}||^2 + \lambda ||\underline{w}||^2$$

This regularization term results in penalization for large $||\underline{w}||^2$. How do we choose this term? depends, there are a lot of algorithms.

6.2.2 Bayesian Regression - Full Solution

The full solution is possible in regression because we are using a distribution with a closed form. The PGM was like to old one, we'll consider the sigma as a given value but we still have the $\underline{w}$ as a random variable.

We can write the Likelihood is the one we've seen before:

$$P(\underline{t} | x, \underline{w}, \tau^2) = N(\underline{t} | x\underline{w}, \sigma^2 \mathbf{1})$$

And then we set the prior on one parameter:

$$P(\underline{w} | \underline{m}_0, s_0) = N(\underline{w} | \underline{m}_0, s_0)$$

The observation here is that this two there are multiplied:

$$N(\underline{t} | x\underline{w}, \sigma^2 \mathbf{1}) \cdot N(\underline{w} | \underline{m}_0, s_0)$$

What we know from the theory is that the posterior will be a Gaussian to (because it's the product of two Gaussian), so we'll have in the end two parameters s_n is the covariant matrix, multiplied by a normalizing constant C for the Gaussian:

$$C \cdot N(\underline{w} | \underline{m}_N, s_N)$$

The technique that we'll use is used for deriving many problem especially in Gaussian, because Gaussian is in a quadratic form. This technique is called completing the square, means that if we have a square of this kind $(a - b)^2$ and assume that a is the unknown parameter (like w) when we develop the square we obtain $a^2 - 2ab + b^2$, so we develop the product of the Gaussian and separate the w : $w^2 - 2aw + b^2$ and b is a constant so we can discard, given that we have two Gaussian we'll develop the quadratic of the other Gaussian and equate the term in which w is doubled and terms in which w is linear.

So in order to have this square term we are taking the log of the posterior, and this is the log of the first normal plus the log of the second normal:

$$\begin{aligned}\log P(\underline{w}|\underline{x}, t, \sigma^2) &\approx \log N(\underline{t}|\underline{w}, x, \sigma^2) + \log N(\underline{w}|\underline{w}, s_0) \\ &= -\frac{1}{2}((\sigma^2 \mathbf{1})^{-1}(\underline{t} - x\underline{w})^T(\underline{t} - x\underline{w}) + (\underline{w} - \underline{m}_0)^T s_0^{-1}(\underline{w} - \underline{m}_0)) + \text{CONST} \\ &= -\frac{1}{2}[\tau^{-2} \underline{t}^T \underline{t} - 2\sigma^{-2} \underline{t}^T x \underline{w} + \underline{w}^T \sigma^{-2} x^T x \underline{w} + \underline{w}^T s_0^{-1} \underline{w} - 2\underline{m}_0^T s_0^{-1} \underline{w} + \underline{m}_0^T s_0^{-1} \underline{m}_0 + \text{CONST}]\end{aligned}$$

Now we have terms that are linear in the w , and it's the mixed term in the quadratic form.

Then we need to identify the quadratic form of the w . And group together the linear and quadratic

$$= -\frac{1}{2}(\underline{w}^T (\sigma^{-2} x^T x + s_0^{-1}) \underline{w} - 2(\sigma^{-2} x^T \underline{t} + s_0^{-1} \underline{m}_0)^T \underline{w})$$

Now we complete the square

$$\begin{aligned}\log N(\underline{w}|\underline{m}_N, s_N) &= -\frac{1}{2}(\underline{w} - \underline{m}_N)^T s_N^{-1}(\underline{w} - \underline{m}_N) + \text{CONST} \\ &= -\frac{1}{2}(\underline{w}^T s_N^{-1} \underline{w} - 2\underline{m}_N^T s_N^{-1} \underline{w} + \underline{m}_N^T s_N^{-1} \underline{m}_N)\end{aligned}$$

$$s_N^{-1} = \sigma^{-2} x^T x + s_0^{-1} \implies \text{covariance matrix} \implies s_N = (\sigma^{-2} x^T x + s_0^{-1})^{-1}$$

$$\underline{m}_N^T s_N^{-1} (\sigma^{-2} x^T \underline{t} + s_0^{-1} \underline{m}_0)^T$$

$$\underline{m}_N^T = s_N (\sigma^{-2} x^T \underline{t} + s_0^{-1} \underline{m}_0) \implies \text{mean}$$

Now that we have the mean and the covariance matrix we have the posterior, because given the mean and the covariance matrix the Gaussian is fully specified:

$$\begin{aligned}P(\underline{w}|\underline{x}, t, \sigma^2) &= N(\underline{w}|\underline{m}_N, s_N) \\ &= C \cdot N(\underline{w}|s_N(\sigma^{-2} x^T \underline{t} + s_0^{-1} \underline{m}_0), (\sigma^{-2} x^T x + s_0^{-1})^{-1})\end{aligned}$$

This provides the posterior:

This shows the advantage of using Gaussian models and it's that we can calculate in the closed form. The mean is influenced by the data point and the weighted prior, and the covariance matrix is a mix between the covariance given by the data plus the covariance of the prior. In the end this posterior of the parameters (different from the prior) will change all the time that we'll see one data. Recall also that when we sample a point from here, this point will be a point of vector (coordinate in this case) w_1, w_2 , this point represent a parameter of the equation of the line $y = w_1 + w_2 x$. When I sample a point basically I'm sampling a vector of parameters so that will be a specific line. So sampling from parameters space means sampling our line. But what happens in practice? the first time we sample a parameter from the prior (because we don't have parameters) and then

we have data point which is sampled given this parameter by our likelihood:

$$\begin{aligned}\underline{w} &\sim P(\underline{w}|m_0s_0) \\ \underline{t}_n &\sim P(\underline{t}_n|w^T x_n, \sigma^2)\end{aligned}$$

Then we read the point, an actual data, what happens is that now we know that this point will change (bias) the Gaussian because we now have a new mean and a new covariance matrix, this will give a posterior and this posterior we'll use for the second point (use the new posterior as the prior to get the data). When I sample from this new condition we sample a line, but the line is constrained to pass through the point

As long we collect data the covariance matrix will restrict, in the end we are sampling a line that is very close to the best line. Reading data we are gaining certainty of where is the line:

6.2.3 Prediction

The second problem that we have is prediction. We have our Bayesian rule:

$$P(\underline{w}|x, \underline{t}) = \frac{P(\underline{t}|x, w)P(\underline{w})}{P(\underline{t}|x)}$$

We distinguish two kind of prediction, the first is the Prior Predictive, the prediction here is given all data and predict a new data, amounts to compute a t new given an x new:

$$\begin{aligned}P(\underline{t}_{new}|\underline{x}_{new}) &= \\ \int P(\underline{t}_{new}|\underline{x}_{new}, \underline{w})P(\underline{w})d\underline{w} \\ \text{this will be a Gaussian}\end{aligned}$$

We have this property of the Gaussian, we have two dimensional Gaussian we can put them together:

Given this property we can rewrite our original problem:

$$N(\underline{t}_{new}|\underline{\eta}_t, \sum_{tt})$$

We know that we obtained this by a linear transformation.

In the final we have the prior predictive distribution:

$$P(\underline{t}_{new}|\underline{x}_{new}) = N(\underline{t}_{new}|\underline{x}_{new}m_0, \underline{x}_{new}^T s_0 \underline{x}_{new} + (\sigma \mathbf{1})^2)$$

Since we have the posterior we can generate prediction using the posterior, this will be a posterior associated with uncertainty. So posterior predictive means that we predict test (or new) data given the louvent posterior $P(\underline{w}|x, t)$.

That means that when we have all data points and we want to predict a new one we also have all the uncertainty associated to this prediction, each new prediction will have uncertainty (it's the square root of the covariance).

In this case we exactly derived the posterior (data and prior to get posterior), once we have the posterior we completed the training part and use this posterior distribution to predict new data.

6.3 Classification

It's a supervised problem. Someone think that classification is a particular form of regression an other theory is based on neuron classification and they think that it's a step to Neural Network.

$$D = \{\underline{x}_n, t_n\}_{n=1}^N, t_n \in \{0, 1\}$$

In this example we consider that in the x we have the square meters of an house and in the y we have the house price, what we aspect here is that the price go higher at the growing of the square meters but there are also other aspect like the house position, that can influence the price. So we can fit a regression model and this would be exactly what we've done for regression. Now we hare somehow interested in determining in the broad distinction high prices and low prices (discretizing the process), the simple way to do this is to set a threshold and using this threshold we might categorize the house price. This operation of putting a threshold make the result non linear because after classification there is no continuity.

To model this general intuition in the first step we have regression, mean that we are considering the likelihood that can be described as usual:

$$P(\underline{t}|\underline{x}, \underline{w}) = \prod_{n=1}^N P(t_n|\underline{z}_n, \underline{w})$$

The PGM is the same:

And the second step is considering that we want discrete output, in this case we have 2 cases to distinguish, from a probabilistic point of view this is just like tossing a coin. T_n is the random variable that take it's realization t_n . We might consider the two probability like the coin toss:

Under this condition we have a Bernoulli distribution (like in coin tossing), that means that the generic random variable T_n will follow the Bernoulli distribution:

$$\begin{aligned} T_n &\sim \text{Bern}(T_n|P) = \\ p^{t_n}(1-p)^{1-t_n} &= \\ p^{t_n}q^{1-t_n} &= \\ P(T_n = 1|\underline{x}_n, \underline{w})^{t_n} P(T_n = 0|\underline{x}_n, \underline{w})^{1-t_n} &= \\ P(\underline{t}|\underline{x}, \underline{w}) \prod_{n=1}^N P(T_n = 1|\underline{x}_n, \underline{w})^{t_n} \cdot P(T_n = 0|\underline{x}_n, \underline{w})^{1-t_n} \end{aligned}$$

This is the Bernoulli model for 2-class classification. Now we have to describe the form of this distribution, for the regression part we simply use the model of regression. Than we need to create some function of a that will transform a in non linearity but that is constraint as the probability belong $(0, 1)$:

$$a = \underline{w}^T \underline{x}_n \implies f(a) \longleftarrow [0, 1]$$

One classical way is to use a function that is usually denoted $\sigma(a)$ that is a sigmoid function (or logistic function):

That can be explicit written:

$$\sigma(a) = \frac{1}{1 + e^{-a}} \in [0, 1]$$

What we are describing is substituting the a in the sigmoid function with the result of the regression part:

$$P(T_1 = 1 | \underline{x}_n, \underline{w}) = \frac{1}{1 + e^{-\underline{w}^T \underline{x}_n}} = \sigma(\underline{w}^T \underline{x}_n)$$

$$P(T_n = 0 | \underline{x}_n, \underline{w}) = 1 - \sigma(\underline{w}^T \underline{x}_n)$$

This function has properties, it's continuous, monotonic increasing, so that means that it's differentiable and the derivative is very simple:

So now we have all tools for trying to infer the parameters of the model. We proceed via maximum likelihood estimation.

6.3.1 MLE for $\underline{w}$

We know the likelihood function (as in regression case):

$$L(\underline{w}) = P(\underline{t} | \underline{x}, \underline{w}) = \prod_{n=1}^N \text{Bern}(t_n | \sigma(\underline{w}^T \underline{x}_n)) = \prod_{n=1}^N \sigma(\underline{w}^T \underline{x}_n)^{t_n} [1 - \sigma(\underline{w}^T \underline{x}_n)]^{1-t_n}$$

Again we'll work in term of log of likelihood:

$$\log L = l(\underline{w}) = \sum_{n=1}^N t_n \log \sigma(\underline{w}^T \underline{x}_n) + (1 - t_n) \log [1 - \sigma(\underline{w}^T \underline{x}_n)]$$

$$z_n = \sigma(\underline{w}^T \underline{x}_n)$$

$$= \sum_{n=1}^N t_n \log z_n + (1 - t_n) \log(1 - z_n)$$

$$\text{Cost of error } J(\underline{w}) = \sum_{n=1}^N t_n \log z_n + (1 - t_n) \log(1 - z_n)$$

This is called cross entropy.

In this case the problem is to find the best $\underline{w}$:

$$\hat{\underline{w}} = \arg \max_{\underline{w}} l(\underline{w}) = \arg \min_{\underline{w}} J(\underline{w}) \implies \Delta_{\underline{w}} J(\underline{w}) = 0$$

$$\underline{\text{delta i.i.d.}} \implies \sum_{i=1}^N (...) \implies \log \prod_{n=1}^N (...) \implies \sum_{n=1}^N (...)$$

$$J(\underline{w}) = \sum_{n=1}^N J_n(\underline{w})$$

So now we have:

$$\begin{aligned}
a_n &= \underline{w}^T \underline{x}_n \Leftarrow \text{activation function} \\
z_n &= \sigma(a_n) \Leftarrow \text{non linearity} \\
J_n(\underline{w}) &= J_n(z_n(a_n(\underline{w}))) \\
\frac{\partial J_n(z_n(a_n(\underline{w})))}{\partial \underline{w}} &= \frac{\partial J_n(z)}{\partial z_n} \cdot \frac{\partial z_n^a}{\partial a_n} \cdot \frac{\partial a_n(\underline{w})}{\partial \underline{w}}
\end{aligned}$$

Now we are in the case of back-propagation:

$$\begin{aligned}
\frac{\partial J_n(z)}{\partial z_n} &= \frac{\partial}{\partial z_n} - [t_n \log z_n + (1 - t_n) \log(1 - z_n)] = -\frac{t_n}{z_n} + \frac{1 - t_n}{1 - z_n} = \frac{-t_n(1 - z_n) + z_n(1 - t_n)}{z_n(1 - z_n)} \\
&= \frac{-t_n + t_n z_n + z_n}{z_n(1 - z_n)}
\end{aligned}$$

Now the second term

So because it's not a linear problem we have to use **gradient descent** (optimization problem) what we'll do for each parameter is something like this:

In this case we have to compute for each cycle the optimization of all data points that can be impossible. We might have another option which is called stochastic gradient descent, we are not doing optimization for all data points but we sample from our data, which means that we are computing for 1 data point:

We have a compromise, which is to use a mini batch of data points:

6.4 Neural Networks

In our classifier context we have a vector x of the features and vector w of weights. We might have the input as a sort of graphical view, this is the input provided to get an activation (it's the deterministic part of the regression) and then this was feed in the logistic function which was named $\int \sigma(a)$ and then we have the output. We might consider all this as a system which we might call (in sort of biological inspiration) the neuron. Actually this regression part develops in a scalar product

If we stick to a specific activation function, we have this formula:

$$y = z(a) = z\left(\sum_{i=1}^D w_i x_i + b\right)$$

If we don't stick to a specific activation function we might have multiple case, the first one is the linear case in which the activation function is the a line:

$$y = z(a) = a$$

An other case is the logistic function in which we have this activation function, the sigmoid (that is used for classification:

$$y = z(a) = \sigma(a)$$

There is also the Tanh (hyperbolic tangent) that is similar to the sigmoid:

$$y = z(a) = \tanh(a)$$

Sometimes the logistic and tanh might collapse in a threshold model:

$$y = z(a) =$$

An other model is the Rectified Linear Unit (ReLU):

$$y = z(a) = a^+ = \max(0, a)$$

6.4.1 Network training

How do we train in order to learn something. In a network perspective, in supervised setting, we have our data, labels $D = \langle \underline{x}_n, t_n \rangle$. The learning is building a mapping from $\underline{x}$ to t (which is the result of the function) $\underline{x} \rightarrow t = y(\underline{x})$.

$$J(w) = d(y_n(\underline{x}; \underline{w}), t_n)$$

$J(w)$ is the cost function (learning is minimize this cost function), t_n are the true values and $y_n(\underline{x}; \underline{w})$ are the model's predictions.

In NN we have advantages: This is a non linear model and if we want to treat this non-linearity mathematically could be a difficult task.

6.4.2 1 Single Neuron

We might have a situation in which we have:

6.5 Back Prop: Multilayer

This is the input that we feed into the network. In general on the generic level l we get the feed from the previous one and add the weights at the same level:

$$\underline{z}^{(l)} = ReLU(w^{(l)} \underline{z}^{(l-1)} + \underline{b}^{(l)})$$

so if we consider in component we find the basic equation for any level:

What we are going to do is to find the equation decomposing the tree parts:

- Final layer that will back propagate
- Go through l level
- Come to the input layer in which we compute the derivative and compute the gradient (correcting errors)

So let's see the final layer first:

We want to find the gradient in respect of the J, we denoted this with $\bar{a}$:

$$\bar{a} \frac{\partial J}{\partial a_j^{(l)}} = \sum_k \frac{\partial J}{\partial a_j^{(l)}} ..$$

We recognize the last term as RELU that is an identity matrix, so basically it's 1. We found the first BP equation.

So we go for the generic layer:

Here we are summing the error computed at the previous layer. At the end we have computed an error at level l, and this final equation can be called the BP(2) equation.

Finally we go to the initial layer, in this layer we have to compute in respect to w_{jk}^l :

And again in this formula we find the error computed at the previous layer. And in the end we found the BP(3) and the BP(4). Now we have all equation.

6.5.1 Matrix notation

In practice we don't use this form of computation, we've seen this example to see what happens at each level, what is done in real software is using the matrix notation. In this notation we need the Hademard or Shun product:

So in this case the BP equations can be compactly written as follows:

No we have all we need, we take the derivative of the final layer and back propagate to the initial layer.

BP algorithm

6.5.2 Optimization

6.5.3 Learning problem